



Energy Conversion and Management

Volume 342, 15 October 2025, 120142

Research Paper

A holistic design framework of centrifugal compressors for high temperature heat pumps considering wet compression prevention and global optimization



Yinlian Yan ^a, Fubin Yang ^{a 1} , Hongguang Zhang ^{a 1}, Nanqiao Wang ^a, Yonghong Xu ^b, Jiabao Cheng ^a, Zhong Ge ^c, Jian Li ^d

[Show more](#)

Outline | Share Cite

<https://doi.org/10.1016/j.enconman.2025.120142>

[Get rights and content](#)

Full text access

Highlights

- A design framework for high temperature heat pump centrifugal compressor is proposed.
- Wet compression of dry fluids is considered at large temperature lifts.

- A global optimization approach coupled with cycle conceptual design is incorporated.
- A systematic case study is performed based on the proposed framework.

Abstract

Centrifugal compressors are an attractive option for high temperature heat pumps (HTHPs) due to advantages such as high efficiency, large capacity, and oil-free operation. Proper design of centrifugal compressors is crucial to realizing the desired HTHP performance. Apart from the strong interactions between different cycle components, there are also complex nonlinear correlations between compressor performance and design parameters. Moreover, the wet compression of dry fluids presents an additional challenge for centrifugal compressors operating at high temperature lifts, further complicating the design of centrifugal compressors for HTHPs. This study proposed a holistic design framework of centrifugal compressors for HTHPs based on the mean-line approach. Real fluid properties and generic design parameters are adopted, considering the flow non-ideality and effective utilization of domain knowledge. A wet compression prevention strategy is introduced to ensure a dry compression by identifying the stage component fluid conditions. Furthermore, a global optimization technique integrated with cycle conceptual design is presented to obtain the optimal compressor design. A case study is performed based on the proposed framework. The influence of design and cycle operating parameters is discussed. A comprehensive comparative assessment on the resultant designs is conducted. Moreover, the corresponding loss distributions are investigated. The case study reveals a clear negative correlation between temperature lift and compressor performance, and diffuser losses exhibit a rapid growth as the condensation temperature increases. This work can provide valuable guidelines for the development of HTHPs with centrifugal compressors.



Keywords

Centrifugal compressor; High temperature heat pump; Mean-line design; Wet compression; Global optimization

Nomenclature

A

cross-sectional area, m²

b

blade height, m

C

absolute velocity, m/s

COP

coefficient of performance

D

diameter, m

D_f

diffusion factor

h

specific enthalpy, kJ/kg

Δh

enthalpy difference

K

correction coefficient

L

length

 $\dot{m}$ molar flow rate, mol/s Ma Mach number N rotational speed, rpm N_s specific rotational speed, rpm P

pressure, kPa

 PR

pressure ratio

 q volumetric flow rate, m³/s $\dot{Q}$

heat flow rate, kW

 R

ratio

 s

specific entropy, kJ/(kg·K)

SF

slip factor

 t

edge thickness, m

 T

temperature, K or °C

 U peripheral velocity, m/s W

relative velocity, m/s

 $\dot{W}$ power consumption, kW Z

number of blades

Greek letters

 α

absolute flow angle, °

 β

relative flow angle, °

 ρ density, kg/m³ τ

blockage factor ψ

tangential velocity coefficient

 ϵ

clearance gap, m

 ϑ angular momentum ξ combination friction factor η

efficiency

 κ inlet hub-to-shroud diameter ratio Ω flow coefficient Φ

work coefficient

Subscripts

0

total

1–6

compressor stations, state points in [Fig. 6](#)

abs

absolute

act

actual

b

blade, back face, blade height

BL

blade loading

c

compressor

CL

clearance loss

cond

condensation

D

diameter

DF

disc friction

diff

diffuser

eva

evaporation

f

fluid

F

full

h

hub

hs

heat source

imp

impellor

inc

incidence

is

isentropic

LL

leakage loss

m

meridional, mean

mix

mixing

out

outlet

rel

relative

RC

recirculation

s

shroud

S

splitter

SF

skin friction

th

theoretical

tt

total-to-total

u

tangential

 ∞

infinite blade hypothesis

Acronyms

HTHPHigh temperature heat pump

IHX

internal heat exchanger

1. Introduction

1.1. Research background

The heat decarbonization in industrial processes, which account for above 20% global CO₂ emissions, is of great significance to achieving the climate goals [[1], [2], [3]]. With the power system transitioning to renewable energy sources, such as solar and wind, generating heat from electricity (also referred to as the electrification of heat) has been recognized as a promising solution to industrial heat decarbonization [[4], [5], [6]].

To promote the electrification of heat in industrial processes, heat pumps (HPs), specifically high temperature heat pumps (HTHPs) are a crucial enabling technology [[7], [8], [9]]. Recent research has suggested that more than half of the industrial heat demand can be satisfied by HTHPs [10]. Moreover, thermodynamically, this technology enables multiple units of useful heat for each unit of electricity, along with the ability to directly recover waste heat from industrial processes to further boost its performance [11]. The energy efficiency or coefficient of performance (COP) can readily exceed 2 with a temperature lift of over 90K [12]. Beyond that, HTHPs also exhibit great potential in offering flexibility to the grid with thermal energy storage combined, which is necessary for high renewable energy penetration [13,14].

1.2. Literature review

Numerous studies have been carried out to unleash the full potential of HTHPs in recent years, where considerable efforts are devoted to cycle configuration, working fluids, and operational optimization. For example, Mateu-Royo et al. [15] evaluated eight HTHP cycle configurations from a comprehensive perspective. Results showed the cascade configuration is the optimal one for temperature lifts above 60K. Tan et al. [16] presented a two-stage evaporation grade compression heat pump system. They found that the novel system is markedly superior than the conventional system with an COP improvement of 1.06–1.29 depending on the operating conditions. Gómez-Hernández et al. [17] proposed a new zeotropic mixture of carbon dioxide and acetone as HTHP working fluids, and reported that a COP of 5.63 is achieved for a 70K temperature difference between the heat sink outlet and heat source inlet. Wu et al. [18] compared the natural refrigerant R718 with other five refrigerants. Results indicated that R718 has better system performance and Carnot efficiency at high temperature outputs. Vannoni et al. [19] conducted a techno-

economic optimization of HTHPs for waste heat recovery, in which the system sizing and component influence are investigated. Huang et al. [20] analyzed the thermodynamic and economic performance of a two-stage CO₂ HTHP. The relationships between system performance and key influencing factors are discussed in their work.

Although significant progress has been made in pushing the limits of the achievable performance of HTHPs in these critical aspects, there is still a strong presumption that the technical challenges from advanced compressors can be addressed [21,22]. The expected compressor for HTHP applications should have the advantages of high efficiency, excellent temperature resistance, large capacity, high pressure ratio operation, and strong reliability. In this context, centrifugal compressors emerge as a highly competitive option for HTHPs [[22], [23], [24]]. Several interesting studies on centrifugal compressors have been reported in the heat pump and refrigeration fields. Schiffmann et al. [25,26] designed and tested a small-scale centrifugal compressor having a 20mm impeller, installed in a domestic heat pump cycle with R134a as working fluids. Results showed the maximal efficiency reaches about 80%. Turunen-Saaresti and Røyttä et al. [27,28] performed off-design performance assessment and operational optimization for a refrigeration cycle with a two-stage centrifugal compressor. Results suggested that the centrifugal compressor also has good operation and efficiency under off-design points. Grauberger et al. [29] experimentally studied a thermally activated cooling system with R1234ze(E), and reported that the centrifugal compressor used can achieve an isentropic efficiency of 84.8% under a pressure ratio of 2.72. Hu et al. [30] developed a centrifugal compressor driven heat pump for district heating. The experimental results showed that the system is quite attractive for district heating with the total COP reaching 4.8. By the way, centrifugal compressors can play an important role in ultra-high-temperature HTHPs operating beyond 160 °C [31], which are essential for further decarbonization of industrial sectors [32], thanks to their capability for oil-free operation. Nevertheless, research on centrifugal compressors for HTHPs is still in its preliminary stage.

For the effective integration of centrifugal compressors in HTHPs, a proper design is indispensable. The performance of centrifugal compressors is strongly affected by working fluids and operational conditions [33,34]. For HTHPs, various potential working fluids can be used [35], which can lead to complex flow phenomena within the compressor [36]. Moreover, HTHPs are typically required to operate under large temperature lifts, high temperature outputs, and large swallowing capacity [12,21]. These can render traditional compressor designs unsuitable for HTHP applications. Therefore, specialized design is essential for HTHP centrifugal compressors. Several studies on this topic have been published. Meroni et al. [36] proposed a mean-line method for the centrifugal compressor design of HTHPs and a coupled optimization strategy with heat pumps. The study suggested conventional enthalpy loss models are applicable to the loss calculation of HTHP centrifugal compressors. Jaatinen-Värri et al. [37] investigated the centrifugal compressor design of a two-stage HTHP and discussed the influence of working fluids

and cycle conditions on the design. It is found that the high Mach numbers could be the main obstacle for high temperature lift. Uusitalo et al. [23] carried out a combined analysis of the cascade HTHP and design of large-scale centrifugal compressors. Results showed that the combination of R245fa (top)/R601 (bottom) produced the best performance for the working fluid combinations evaluated. In another work [38], they studied the centrifugal compressor design with hydrocarbons installed in large-scale HTHPs, and reported that cycle components can make a notable difference in the compressor design, and the cyclic hydrocarbons resulted in a superior COP. Jiang et al. [39] introduced the design of a back-to-back centrifugal compressor used in carbon dioxide HTHPs with an axial force optimization method developed. The influence of fluid inlet conditions, bearing stiffness, and the motor rotor diameter were discussed in the study.

1.3. Research gaps

According to the literature, existing efforts primarily focus on the design method and the combined analysis of compressor design and heat pump thermodynamics under specific working fluids, cycle configurations, and applications. There are still clear gaps need to be addressed. The first is that current design methods reported are typically developed from air centrifugal compressors [23,36], [37], [38]]. Although the suitability of these methods has been validated, they failed to give sufficient consideration to the complex thermodynamic behavior of fluids in HTHP applications, which can further complicate centrifugal compressor design. Specifically, many working fluids recommended for HTHPs, such as R1336mzz(Z), R1233zd(E), and cyclopentane, feature with a positive slope of saturation vapour curve ($ds/dT > 0$), which is referred to dry fluids. Due to this characteristic, there is an inherent risk of wet compression (entering the two-phase region in compression), especially at high temperature lifts [12], which can incur mechanical damage and efficiency deterioration of centrifugal compressors [23,40]. Even if reasonable superheating at the compressor entrance is ensured, this still necessitates additional considerations in HTHP centrifugal compressor design since the geometric and operational designs are strongly coupled with fluid states [41,42].

Furthermore, the literature review suggests that systematic approach for optimization at the early design stage is still lacking. Current design methods usually adopt a sequential procedure, in which the desired compression efficiency is assumed firstly to generate design inputs for the compressor design model, and the design process is terminated once a feasible design is achieved. Several limitations exist for such approaches. Firstly, this can lead to design failure if overly optimistic assumptions are made, leading to a more complicated and time-consuming design process. Additionally, the design procedure can yield a suboptimal design if the efficiency is underestimated [43]. This, in turn, can also badly impact the performance assessment at the system level. On the other hand, existing research often relies on certain predefined compressor design variables based on experience, or

adopts non-generic design variables (e.g., rotational speed). The former ignores the synergistic effects of parameters, while the latter complicates the determination of an appropriate decision scope, making the design and optimization process more challenging.

1.4. Work and innovations

To close the research gaps, this study proposed a holistic design framework of centrifugal compressors for HTHPs based on the mean-line approach. In this framework, the flow non-ideality is accounted by incorporating real gas properties, and generic design parameters are applied to enable the effective utilization of domain design experience. Additionally, a strategy is incorporated to avoid the potential wet compression issue at large temperature lifts when dry fluids are used, by identifying the fluid conditions of key stage components. A global search algorithm is developed, accounting for the complex nonlinearity between design variables and performance. The compressor design is directly integrated with the cycle conceptual design of HTHPs considering the strong coupling between the cycle and components. A case study is carried out to evaluate and analyze the optimal centrifugal compressor designs and cycle performance. The effects of key design and cycle operating parameters are discussed. A comprehensive comparative assessment on the resultant design schemes is also performed using the design characterization parameters of centrifugal compressors, along with a benchmarking study. In addition, the irreversible losses distributions in the compressor are also investigated.

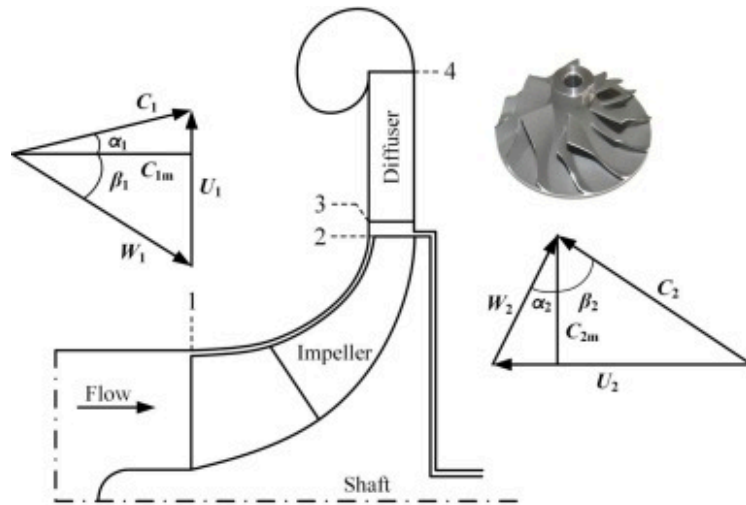
Compared to prior research, the major novelties and advancements of this work can be summarized in three main areas. Firstly, the complexity associated with common HTHP working fluids are taken into account by incorporating a wet compression prevention strategy. Secondly, a holistic design concept is introduced, in which compressor design and HTHP cycle design are coupled together to yield an overall optimal design scheme. Lastly, the proposed framework enables an effective and convenient determination of the design variable decision space by adopting generic design variables, whose decision boundaries can be determined based on the existing domain knowledge. This work can provide systematic design guidelines for the future development of HTHPs with centrifugal compressors, paving the way for their more efficient and reliable application in industrial processes.

2. Methodology

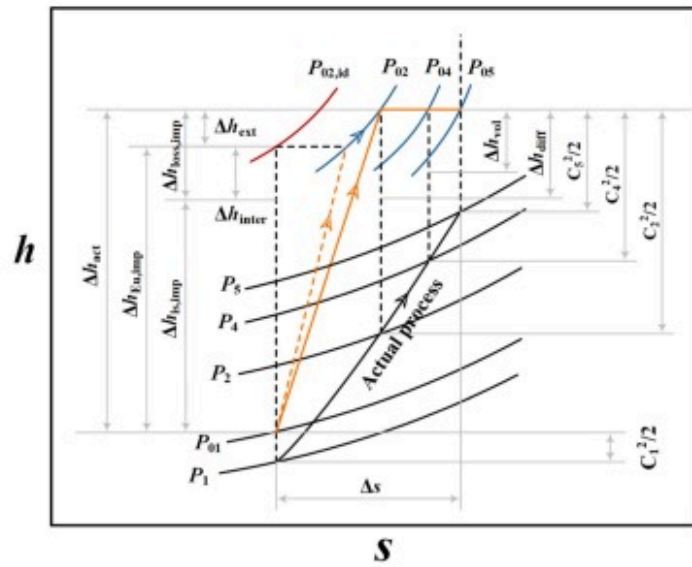
In this section, the proposed design framework is systematically presented. In the following, the detailed compressor aerodynamic design modeling, strategy for the wet compression prevention, global optimization technique integrated with cycle conceptual design, and model validation are provided, respectively.

2.1. Centrifugal compressor aerodynamic design modeling

Centrifugal compressors mainly consist of the rotating impeller, stationary diffuser, and volute. During compression, the impeller first imparts kinetic energy to the flow, then the diffuser converts the kinetic energy into the pressure energy, and finally the volute smoothly discharges the flow. The structural schematic of centrifugal compressors and corresponding h - s diagram is shown in [Fig. 1](#).



(a)



(b)

[Download: Download high-res image \(236KB\)](#)

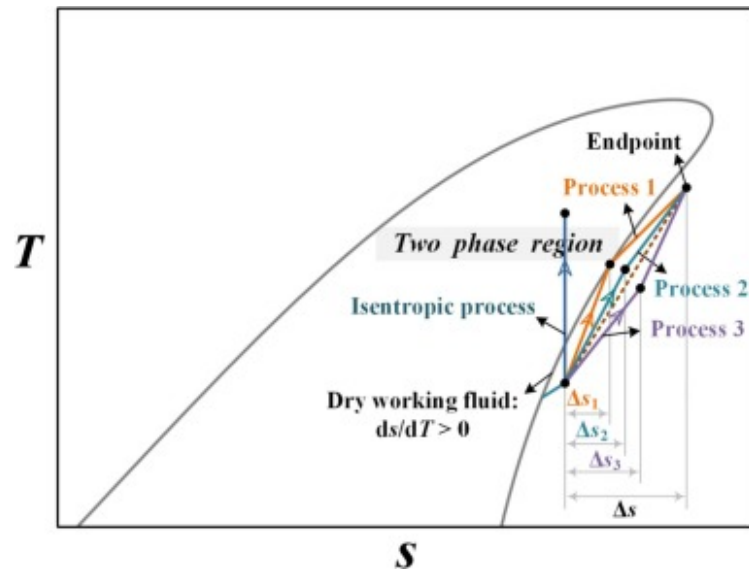
[Download: Download full-size image](#)

Fig. 1. Schematic of centrifugal compressors (a) and corresponding h - s diagram (b).

2.1.1. Considerations for high-temperature heat pump centrifugal compressor design

Accurate fluid properties are essential for the optimal design of HTHP centrifugal compressors due to the strong coupling between fluid thermodynamics and compressor performance [23]. In classical application-specific design approaches, the ideal gas equation of state (EOS) is often used to calculate fluid properties [44]. However, for the HTHP centrifugal compressor, the working fluid typically operates near saturated gas states, where significant non-ideal behavior emerges during compression [44,45]. This deviation renders the ideal gas approximation invalid. Therefore, real gas EOSs must be incorporated to provide accurate fluid property predictions.

Another key aspect in HTHP centrifugal compressor design is to consider the risk of wet compression, which can bring mechanical damage and significantly reduce compressor efficiency [12]. Wet compression becomes particularly relevant in HTHP applications due to the combination of high-temperature lift requirements and the thermodynamic behavior of potential working fluids such as R1336mzz and R1233zd [12,15]. These fluids, classified as dry fluids with a positive slope on their saturation vapor curve in the T - s diagram, tend to approach or enter the two-phase region during compression at elevated temperature lifts, as shown in Fig. 2. Apart from maintaining sufficient superheat at the compressor inlet [12], the reasonable compressor design also plays a key role in avoiding wet compression. As illustrated by Fig. 2, even for the same compression endpoint, the thermodynamic trajectory of the working fluid during the compression process can be very different (see processes 1, 2, and 3). This discrepancy in compression trajectory directly linked to the structural characteristics of compressor stage components. Hence, a focus in the development of HTHP centrifugal compressor design models should be put on the wet compression prevention.



[Download: Download high-res image \(100KB\)](#)

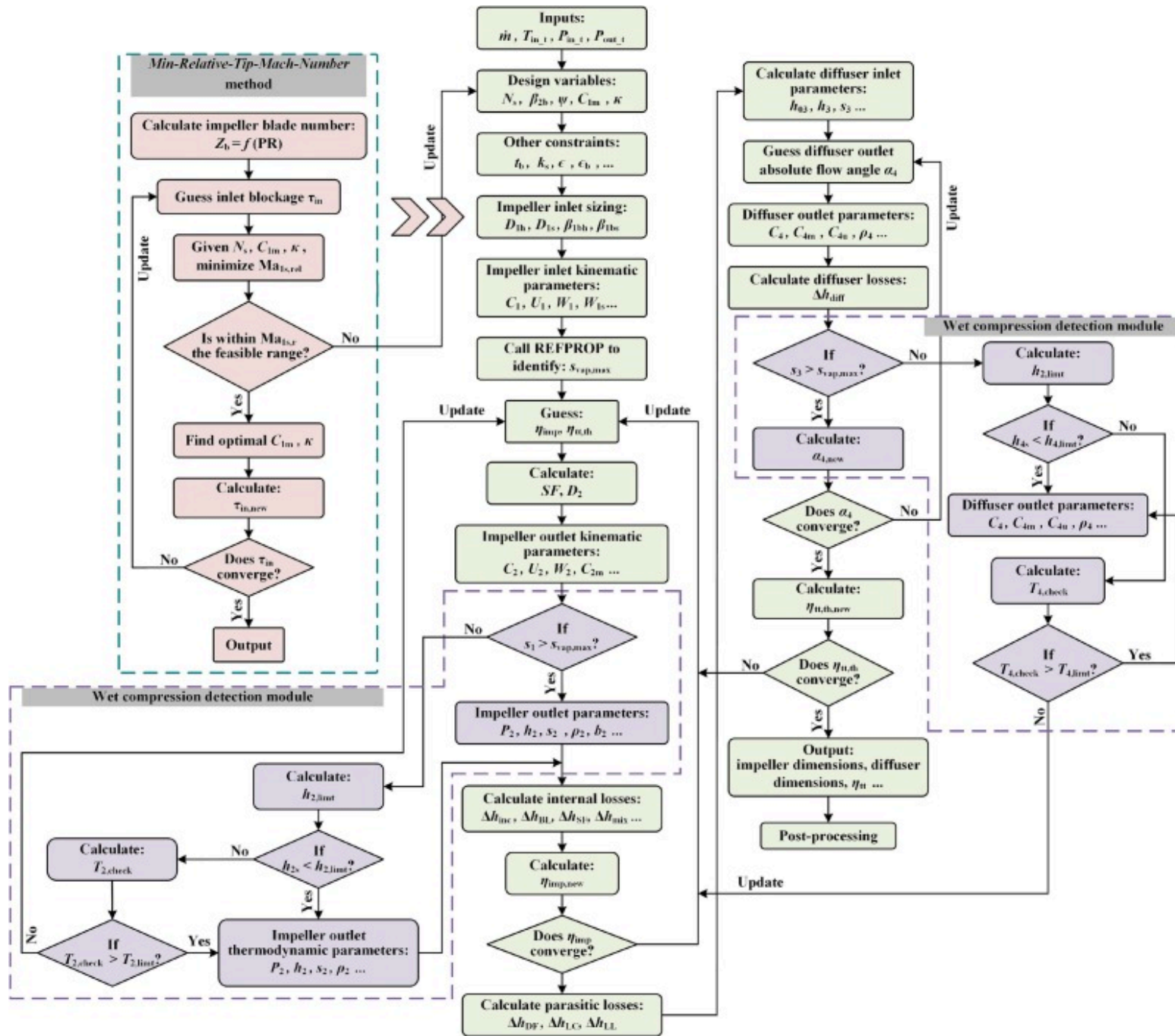
[Download: Download full-size image](#)

Fig. 2. Illustration of the fluid thermodynamic trajectories during compression.

2.1.2. Real gas-based design framework considering wet compression prevention

The proposed design framework is based on the classical mean-line method of the turbomachinery, which is inherently physics-based and mechanism-based [36]. The fundamental concept behind is to construct the correlation between the dimensional parameters and compressor performance at specific rotational speed and design conditions. This is followed by the solving of the kinematic and thermodynamic parameters on the key meridional stations of a radial stage by the mass continuity, energy and angular momentum conservation, loss correlations, and equations of state (EOSs). It should be noted that the scalability to higher-temperature applications is a key aspect for the practical value of the proposed framework. With the non-ideal thermodynamic behavior of working fluids described accurately, the proposed framework can, in principle, be extended to higher-temperature applications. Nevertheless, challenges associated with ultra-high-temperature materials must first be addressed.

To account for the non-ideal behavior of HTHP working fluids, the presented design framework utilizes the real gas EOSs from REFPROP to obtain accurate fluid properties [46]. Additionally, to prevent the possible wet compression issue, a strategy is incorporated to avoid improper stage component geometry in the design framework. Fig. 3 illustrates the systematic design framework of HTHP centrifugal compressors considering wet compression prevention.



Download: [Download high-res image \(781KB\)](#)

[Download: Download full-size image](#)

Fig. 3. Design framework with wet compression prevention strategy for HTHP centrifugal compressors.

The design begins with inputs derived from upstream and downstream cycle components, including the mass flow rate, inlet total temperature and pressure, and outlet total pressure. The design variables that determine compressor geometry are the specific speed N_s , tangential velocity coefficient ψ , impeller outlet blade angle β_{2b} , inlet meridional velocity C_{1m} , inlet hub-to-shroud diameter ratio κ . These design variables have clear physical meanings and the advantage of generalizability for compressors of different sizes [47]. Additionally, several design constraints, such as blade thickness and impeller clearances, are required, which mainly depend on current manufacturing levels.

2.1.2.1. Impeller inlet design

To size the impeller inlet, the Min-Relative-Tip-Mach-Number method is used, which is important to ensure the high design point efficiency of centrifugal compressors [43,48,49]. The core idea is to minimize the relative Mach number at the inlet shroud $Ma_{1,rel}$. The intrinsic motivation involves two aspects. The first is to reduce the incidence losses and frictional losses along the impeller flow path, and the second is to enlarge the stable operating range between choke and stall. To start, the number of impeller blades is first determined by the following empirical correlations [50]:

$$Z_b = \begin{cases} 12.03 + 2.544PR_{tt} & \text{(without splitters)} \\ -4.527e^{1.865/PR_{tt}} + 32.22 & \text{(with splitters)} \end{cases} \quad (1)$$

where PR_{tt} is the stage total-to-total pressure ratio.

The impeller rotational speed is specified by the specific speed, defined as:

$$N_s = Nq_{in}^{1/2} / \Delta h_{is,tt}^{3/4} \quad (2)$$

where N is the rotational speed, q_{in} is the volumetric flow rate at the compressor inlet, and $\Delta h_{s,tt}$ is the isentropic compression enthalpy.

Based on the mass conservation at the impeller inlet station, the following relation is established:

$$m_f = \rho_1 \pi C_1 D_{1,eq} \cos(\alpha_1) (1 - \tau_{in}) / 4 \quad (3)$$

where ρ_1 is the static density, C_1 and α_1 is the absolute velocity and absolute flow angle at the inlet station, $D_{1,eq}$ is the equivalent diameter, and τ_{in} represents the inlet blockage factor. The case of swirl-free inflow is assumed, and thus α_1 is zero in this study.

The equivalent diameter $D_{1,eq}$ and blockage factor τ_{in} is defined as:

$$D_{1,eq} = (D_{1s}^2 - D_{1h}^2)^{1/2} \quad (4)$$

$$\tau_{in} = 4Z_{b1} t_{b1} (D_{1s} - D_{1h}) / (\pi (D_{1s}^2 - D_{1h}^2)) = 4Z_{b1} t_{b1} / (\pi (D_{1s} + D_{1h})) \quad (5)$$

where t_{b1} is the leading edge thickness, and D_{1s} and D_{1h} is the impeller inlet diameters at the shroud and hub, respectively.

Using the preset inlet hub-to-shroud diameter ratio κ , D_{1s} and D_{1h} can be calculated as:

$$D_{1s} = (D_{1,eq}^2 / (1 - \kappa))^{1/2} \quad (6)$$

$$D_{1h} = \kappa D_{1s} \quad (7)$$

With the rotational speed and inlet shroud diameter, the peripheral speed at the shroud is given as:

$$U_{1s} = \pi N D_{1s} / 60 \quad (8)$$

According to the inlet velocity triangle, the relative tip Mach number at the shroud is calculated as:

$$Ma_{1s,rel} = ((U_{1s} - C_1 \sin(\alpha_1))^2 + (C_1 \cos(\alpha_1))^2)^{1/2} / C_{1,sound} \quad (9)$$

where $C_{1,sound}$ is the speed of sound at the inlet station and computed by using REFPROP:

$$C_{1,sound} = C_{sound}(h_1, s_1) \quad (10)$$

Once the above mathematical relationships are established, the following optimization problem can be formulated based on the Min-Relative-Tip-Mach-Number method.

$$\begin{cases} \text{Minimize} & Ma_{1s,rel} (C_1, \kappa) = ((U_{1s} - C_1 \sin(\alpha_1))^2 + (C_1 \cos(\alpha_1))^2)^{1/2} \\ & / C_{sound}(h_1, s_1) \\ \text{Subject to} & C_1 \subseteq R_1, \kappa \subseteq R_2 \end{cases} \quad (11)$$

where R_1 and R_2 correspond to the range of decision variables.

To account for the impeller blade effect on the actual effective area at the inlet station, a calibration process is performed through iteration. The guessed τ_{in} is updated by Eq. (5) until the convergence criterion is satisfied. Moreover, to reduce the unnecessary design failure and computational cost, an strategy based on the feasible range of $Ma_{1,rel}$ and $Ma_{1s,rel}$ is employed [51]. That is, when the resulting $Ma_{1,rel}$ exceeds the feasible range, the design variables are adjusted accordingly.

Under the minimum $Ma_{1,rel}$, the impeller inlet blade angles are further determined based on the zero incidence loss rule [49]. The blade angles at the hub and shroud are calculated as:

$$\beta_{1b,h} = \arctan(A_1 / A_{1,th} \tan(\beta_{1h})) \quad (12)$$

$$\beta_{1b,s} = \arctan(A_1 / A_{1,th} \tan(\beta_{1s})) \quad (13)$$

2.1.2.2. Impeller outlet design

For the determination of the impeller outlet geometry, the assumptions on impeller and stage efficiencies are needed. Hence, the design involves multiple iterations. The design variables used are the impeller outlet blade angle β_{2b} and tangential velocity coefficient ψ . To obtain other dimensions, the kinematic and thermodynamic parameters at the outlet station need to be solved. Firstly, according to the momentum conservation, the theoretical specific work done by the impeller to the fluid is calculated as:

$$\Delta h_{Euler} = h_{02} - h_{01} = C_{2u} U_2 - C_{1u} U_1 \quad (14)$$

where h_{01} and h_{02} is the total enthalpy at the impeller inlet and outlet, respectively.

By assuming an isentropic compression process, the actual enthalpy increase can be written as:

$$h_{02} - h_{01} = \Delta h_{is,tt} / \eta_{tt,th} \quad (15)$$

where $\eta_{tt,th}$ is the theoretic isentropic efficiency of the stage.

Thereby, the total specific enthalpy at the outlet station for a given $\eta_{tt,th}$ is:

$$h_{02} = h_{01} + \Delta h_{\text{Euler}} = h_{01} + \Delta h_{is,tt} / \eta_{tt,th} \quad (16)$$

To calculate the kinematic parameters, the velocity slip phenomenon at impeller outlet should be accounted. This leads to the calculation of the slip factor, which is given by:

$$SF = 1 - \Delta C_{2u} / U_2 \quad (17)$$

$$= \begin{cases} SF & \sigma < \sigma_{\text{lim}} \\ SF(1 - ((\sigma - \sigma_{\text{lim}}) / (1 - \sigma_{\text{lim}}))^{((90^\circ + \beta_{2b})/10)^{1/2}}) & \sigma > \sigma_{\text{lim}} \end{cases}$$

$$SF = 1 - \cos(\beta_{2b})^{1/2} / Z_b^{0.7} \quad (18)$$

$$\sigma_{\text{lim}} = (D_1 / D_2)_{\text{lim}} = \sin(19^\circ + 0.2(90^\circ - \beta_{2b})) \quad (19)$$

where σ is the diameter ratio of the impeller inlet and outlet.

Based on SF and ψ , the peripheral velocity at the outlet is solved as:

$$\psi = C_{2u,\infty} / U_2 \quad (20)$$

$$U_2 = (\Delta h_{\text{Euler}} / (\psi - (1 - SF)))^{1/2} \quad (21)$$

Accordingly, the impeller outlet diameter can be determined by:

$$D_2 = 60U_2 / \pi / N \quad (22)$$

From the outlet velocity triangle constructed under the infinite blade hypothesis, the absolute tangential velocity can be expressed as:

$$C_{2u,\infty} = U_2 - C_{2m} \tan(\beta_{2b}) \quad (23)$$

Using the trigonometry, the absolute velocity at the outlet station is given by:

$$C_2 = (C_{2m}^2 + C_{2u}^2)^{1/2} \quad (24)$$

The static enthalpy is solved from the total-to-static relation:

$$h_2 = h_{02} - C_2^2/2 \quad (25)$$

By defining the impeller isentropic efficiency η_{imp} , the isentropic static enthalpy is obtained:

$$h_{2,is} = h_1 + (h_2 - h_1) \eta_{imp} \quad (26)$$

After that, other thermodynamic parameters can be evaluated from REFPROP:

$$P_2 = P(h_{2,is}, s_1) \quad (27)$$

$$s_2 = s(P_2, h_2) \quad (28)$$

$$\rho_2 = \rho(P_2, h_2) \quad (29)$$

where P_2 , s_2 , ρ_2 are the static pressure, entropy, and density, respectively.

By using the continuity equation at the outlet station, the blade height is obtained:

$$b_2 = \dot{m}_f / (\rho_2 (\pi D_2 - Z_b t_2) / C_{2m}) \quad (30)$$

where t_2 is the trailing edge thickness.

To derive the geometry of the impeller outlet, an iteration of η_{imp} is followed, which depends on the internal losses of the impeller. With all thermodynamic and kinematic parameters at the impeller inlet and outlet established, the internal losses can be calculated, followed by a update of the estimation of η_{imp} . Once η_{imp} converges, the accurate impeller outlet conditions are obtained.

According to the loss mechanism, the impeller internal losses can be divided into the incidence loss (Δh_{inc}), blade loading loss (Δh_{BL}), skin friction loss (Δh_{SF}), clearance loss (Δh_{CL}), mixing loss (Δh_{mix}) and choking loss (Δh_{choke}). The detailed calculation

methods of these internal losses are listed in Appendix A.

Once all the internal losses are solved, the total static enthalpy can be obtained:

$$h_2 = h_1 + \Delta h_{\text{inc}} + \Delta h_{\text{BL}} + \Delta h_{\text{SF}} + \Delta h_{\text{CL}} + \Delta h_{\text{mix}} + \Delta h_{\text{choke}} \quad (31)$$

and then the impeller efficiency is updated accordingly.

Besides the internal losses, there are also parasitic losses for the impeller, which account the extra energy consumed and influence the fluid conditions in the stators. These losses include disc friction loss (Δh_{DF}), recirculation loss (Δh_{RC}), and leakage loss (Δh_{LL}). The detailed calculation methods of these parasitic losses are listed in Appendix A.

With all the parasitic losses calculated, the total enthalpy at the diffuser inlet can be determined:

$$h_{03} = h_{02} + \Delta h_{\text{DF}} + \Delta h_{\text{RC}} + \Delta h_{\text{LL}} \quad (32)$$

2.1.2.3. Diffuser design

After leaving the impeller, the fluid still has considerable amount of kinetic energy. The diffuser is used to recover part of the kinetic energy as pressure energy. In this study, the vaneless diffuser is considered. The main structural parameters of the vaneless diffuser are height and the diameters at the inlet and outlet. These structural parameters are chosen according to the impeller dimensions. Subsequently, the outlet conditions are determined using the approach from [47]. Firstly, the angular momentum of the flow in the diffuser needs to be defined:

$$\vartheta = C_u D/2 \quad (33)$$

The variation of the angular momentum along the radius complies with the following relation:

$$\vartheta = \vartheta_i \exp(\xi) \quad (34)$$

where ξ is the combination friction factor, and is calculated as:

$$\xi = -0.5K_f D_{i-1} (D_i/D_{i-1} - 1) / b_m / \sin(90^\circ - \alpha_m) \quad (35)$$

where ξ is the combination friction factor, b_m is the mean height, and α_m is the mean absolute flow angle.

thereby, the absolute tangential velocity at diffuser outlet is:

$$C_{4u} = C_{2u} D_2 \exp(\zeta_3) / D_4 \quad (36)$$

Based on the velocity angle and continuity equation at the outlet, the diffuser outlet density is solved by:

$$\rho_4 = m_f / (A_4 C_{4u} \tan(90^\circ - \alpha_4)) \quad (37)$$

By using an estimate of α_4 , the diffuser outlet conditions and corresponding irreversible losses can be solved:

$$\Delta h_{\text{diff}} = K_f (D_4 - D_2) (C_2^2/b_2 + C_4^3/b_4) / \sin(90^\circ - \alpha_m) \quad (38)$$

After that, an iterative verification of the outlet density is required by using:

$$\rho_4 = \rho(P_4, h_4) \quad (39)$$

With all the losses are evaluated, it is possible to calculate the new $\eta_{\text{tt,th}}$ by:

$$\eta_{\text{tt,th}} = (h_{02} - \Delta h_{\text{inc}} - \Delta h_{\text{BL}} - \Delta h_{\text{SF}} - \Delta h_{\text{CL}} - \Delta h_{\text{mix}} - \Delta h_{\text{choke}} - \Delta h_{\text{diff}}) / (h_{02} - h_{01}) \quad (40)$$

Once the new $\eta_{\text{tt,th}}$ is obtained, the prior estimate is then updated for another iteration. When the convergence of $\eta_{\text{tt,th}}$ is achieved, the actual efficiency of the compressor can be obtained by:

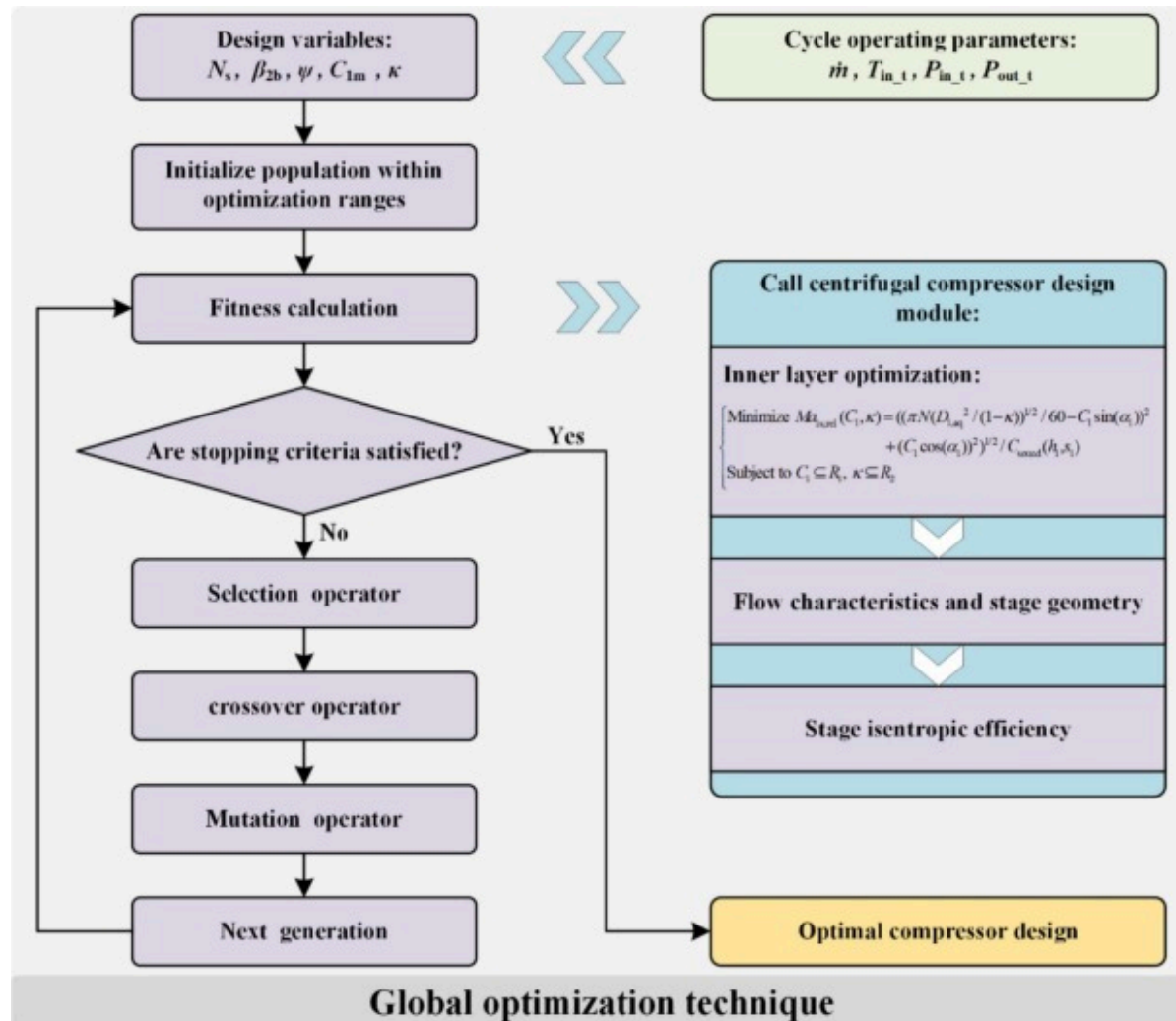
$$\begin{aligned} \eta_c &= \Delta h_{\text{is,tt}} / (h_{01} + \Delta h_{\text{inc}} + \Delta h_{\text{BL}} + \Delta h_{\text{SF}} + \Delta h_{\text{CL}} + \Delta h_{\text{mix}} + \Delta h_{\text{choke}} + \Delta h_{\text{diff}} \\ &\quad + \Delta h_{\text{DF}} + \Delta h_{\text{RC}} + \Delta h_{\text{LL}}) \end{aligned} \quad (41)$$

2.1.2.4. Wet compression prevention strategy

As illustrated above, the improper compressor design can lead to the undesired wet compression when the dry fluid is applied for HTHP applications. Therefore, a wet compression prevention strategy is introduced in the design framework. The core idea is to identify the compression endpoints for the stage components. The corresponding procedure is displayed in Fig. 3. For any given working fluid, the maximal specific entropy on the saturation vapour curve $s_{\text{vap,max}}$ is first calculated based on the design operating conditions of HTHP. Then, the inlet entropy of the stage components $s_{\text{sc,in}}$ is compared with $s_{\text{vap,max}}$ to evaluate the possibility of wet compression. For the case that $s_{\text{sc,in}}$ is less than $s_{\text{vap,max}}$, there are the potential risk of wet compression, and then it moves to calculate the isentropic compression enthalpy for further comparison with its upper limit $h_{\text{is,limit}}$. If the upper limit is exceeded, the influence of irreversibility is accounted. This follows the superheating judgement. For the case the superheating is not satisfied, the design variables are adjusted accordingly. Thereby, the potential wet compression can be effectively avoided at the design phase.

2.2. Optimization technique

Due to the strong nonlinearity of design parameters and performance, global searching within the design space is necessary to realize the optimal performance of centrifugal compressors. Moreover, the integration of cycle conceptual design and component design is required to avoid the suboptimal solutions and design failure. For these reasons, a genetic algorithm based global optimization technique is presented, which is widely used for such problems [36,52]. The corresponding concept is shown in Fig. 4. Firstly, the cycle design parameters are given to generate the required design inputs of centrifugal compressors, namely, fluid states at the compressor inlet, mass flow rate, and the outlet pressure. After that, the design variables are initialized within the design space. Owing to the adoption of generic design variables for which the decision boundaries can be determined by the existing domain knowledge [47,53], the design space generated is capable of achieving an efficient and successful design. Then, the fitness of the possible solutions, namely, the isentropic efficiency, is calculated through the design model. In this part, an embedded sub-optimization is used based on the Min-Relative-Tip-Mach-Number method to ensure the efficient design. Subsequently, a selection operator with elitist preservation is employed to generate individuals for further processing by the crossover and mutation operators, in which random next generation are produced. By alternately using the fitness calculation and the three operators, the global optimization can be realized.



[Download: Download high-res image \(162KB\)](#)

[Download: Download full-size image](#)

Fig. 4. Global optimization technique for HTHP centrifugal compressor design.

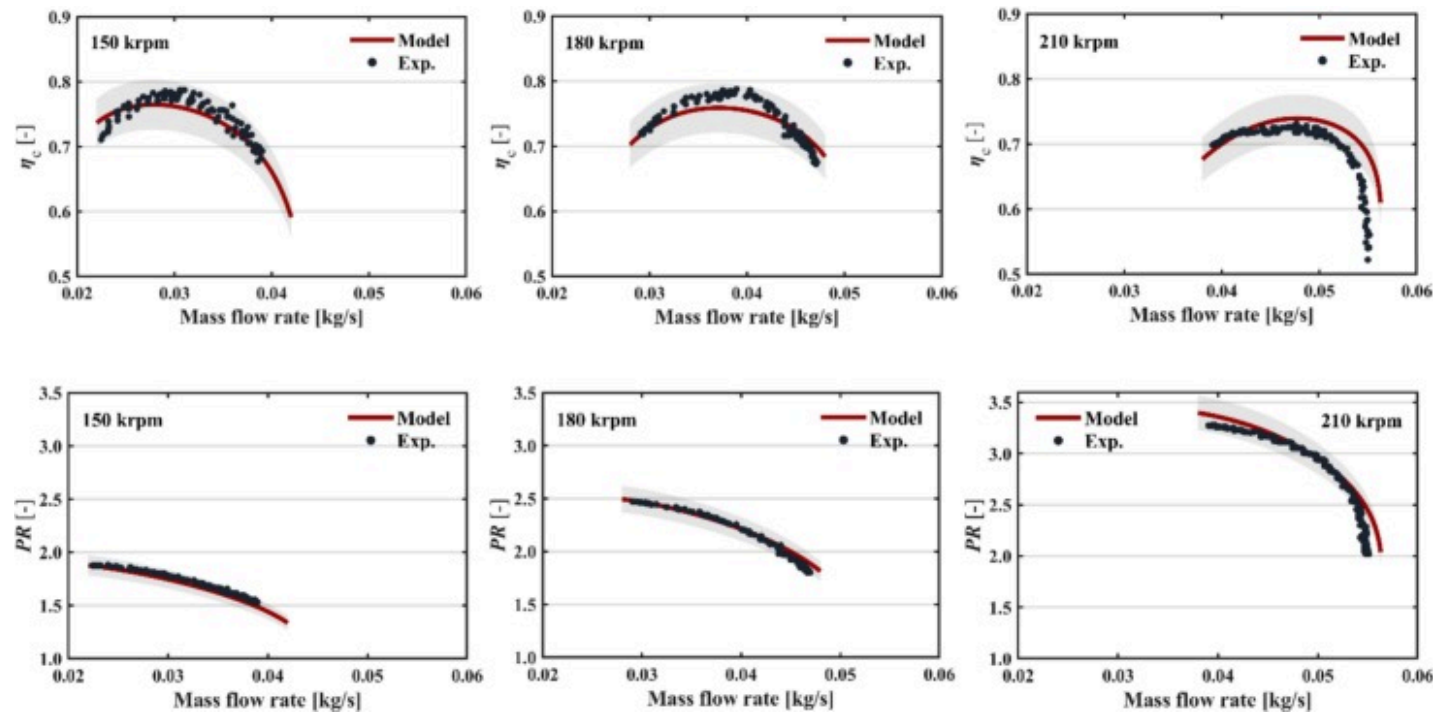
2.3. Validation

The presented compressor design framework follows the classical meanline design approach of centrifugal compressors [33,34,41]. Compared with the previous studies [23,24], the proposed compressor design methodology shares the same fundamental physical principles at its core, which should guarantee its theoretical reliability. Moreover, the adopted loss correlations are all mechanism-based, and the vast majority of them have been employed in the geometrical design and off-design performance prediction of HTHP centrifugal compressors [23,34]. Nevertheless, a validation is performed to ensure the reliability. The proposed design methodology is validated by comparison with well-documented centrifugal compressor test cases, given that the essence of the validity lies in the ability to provide reliable performance evaluations under given boundary conditions. The experimental results of a centrifugal compressor designed for domestic heat pump applications are used [25,26]. The specifications of the compressor and test conditions are listed in Table 1. This validation test case was selected primarily because, to our best knowledge, it is the only publicly available and well-documented source of experimental data on heat pump centrifugal compressors. Moreover, previous research [23,34] has also adopted this test case for model validation. The design model is adjusted accordingly to offer the corresponding performance predictions. Fig. 5 displays the comparisons of the pressure ratio (PR) and isentropic efficiency between the experimental data and the predictions. As illustrated by the figure, the model correctly captures the trend of η_c and PR . Moreover, good prediction accuracy is achieved with the vast majority of the predictions in the $\pm 5\%$ uncertainty bands, except the region where the performance deteriorates rapidly at the rotational speed of 210krpm. Hence, the validity of the model can be guaranteed.

Table 1. Specifications of the compressor and test conditions in Refs. [25,26].

Variable	Unit	Value
Working fluid	–	R134a
Rotational speed	krpm	150~210
Inlet total temperature	K	283.15
Inlet total pressure	bar	1.65
Impeller inlet diameter at the shroud	mm	11.2
Impeller inlet diameter at the hub	mm	2

Variable	Unit	Value
Impeller outlet diameter	mm	20
Impeller inlet blade angle at the shroud	°	−20.5
Impeller inlet blade angle at the hub	°	−60
Impeller outlet blade angle	°	45
Impeller outlet blade height	mm	1.2
Diffuser outlet diameter	mm	33
Diffuser outlet diameter	mm	1
Splitter blades	–	9
Full blades	–	9
Impeller axial length	mm	7.7
Leading edge thickness	mm	0.2
Trailing edge thickness	mm	0.2
Surface roughness	mm	0.01
Clearance between the casing and the impeller	mm	0.05
Impeller back face clearance	mm	0.2



[Download: Download high-res image \(256KB\)](#)

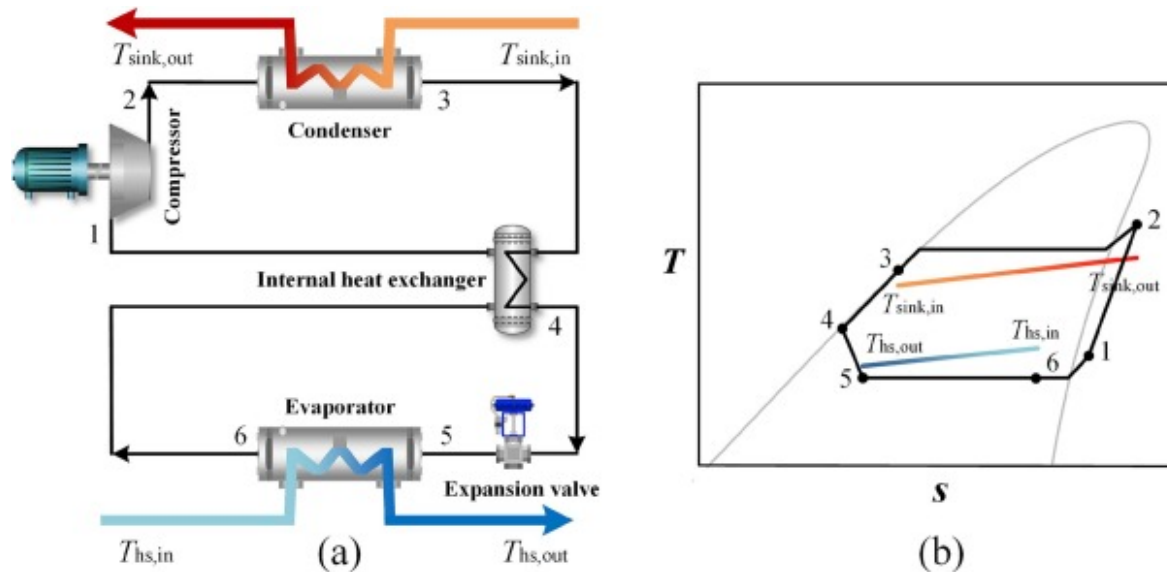
[Download: Download full-size image](#)

Fig. 5. Comparison of compressor efficiency and pressure ratio with the experimental results of the compressor with R134a from Refs. [25,26]. The shaded area represents the $\pm 5\%$ uncertainty band.

3. Case study and modeling

In this section, the proposed framework is applied for the design analysis of a single-stage HTHP driven by centrifugal compressors. The corresponding configuration and T - s diagram are shown in Fig. 6. The main components of the system include the evaporator, compressor, condenser, expansion valve, and internal heat exchanger (IHX). During the operation, the working fluid first receives heat from a heat source in the evaporator (5–6), and then flows into the IHX to be superheated by the working fluid after the condensation (6–1). Subsequently, it is pressurized by the compressor to a high-temperature and high-pressure

state (1–2), and later enters the condenser and releases heat to the heat sink (2–3). Then, the working fluid is further cooled in the subcooler (3–4), and finally expands to the initial state through the expansion valve to close the cycle.



[Download: Download high-res image \(186KB\)](#)

[Download: Download full-size image](#)

Fig. 6. Configuration of the HTHP system and corresponding T - s diagram.

To perform the design analysis, the thermodynamic modeling of the HTHP system is firstly conducted. The modeling process begins with the commonly used assumptions:

- (1) The system operates in a steady state;
- (2) The heat losses and pressure drops are ignored;
- (3) The gravitational potential energy and kinetic energy of the working fluid are negligible.

The main modeling logic is as follows:

By using energy balance in the evaporator, the mass flow rate of working fluid is calculated as:

$$\dot{m}_f = \dot{Q}_{hs} / (h_6 - h_5) \quad (42)$$

where $\dot{Q}_{hs}$ is the heat source capacity, h is the specific enthalpy of the working fluid, and the subscripts represents the states point in Fig. 6.

The energy conservation equation in the IHX is:

$$\dot{m}_f (h_3 - h_4) = \dot{m}_f (h_1 - h_6) \quad (43)$$

The power consumption of the compressor is obtained from:

$$\dot{W}_C = \dot{m}_f (h_2 - h_1) / \eta_{el} = \dot{m}_f (h_{2s} - h_1) / \eta_c / \eta_{el} \quad (44)$$

where η_{el} is the electrical efficiency of the compressor, which is assumed as 0.98, and η_c is the isentropic efficiency.

The heating power of the HTHP system is calculated as:

$$\dot{Q}_{cond} = \dot{m}_f (h_2 - h_3) \quad (45)$$

The pressure ratio of the compressor is given by:

$$PR = P_{cond} / P_{eva} \quad (46)$$

where P_{cond} and P_{eva} are the condensation and evaporation pressures, respectively.

The key performance indicators of HTHP, COP and volumetric heating capacity (VHC), can be calculated as:

$$COP = \dot{Q}_{cond} / \dot{W}_C \quad (47)$$

$$VHC = (h_2 - h_3) / v_1 \quad (48)$$

To analyze the effects of design variables, the sensitivity coefficient [54] is adopted and defined as

$$S_{BF} = |(\Delta B/B) / (\Delta F/F)| \quad (49)$$

Table 2 lists the operating parameters of the HTHP system. The industrial waste heat is considered as the heat source, with the temperature assumed to be 80 °C. The condensation temperature varies from 100 °C to 130 °C in order to investigate the influence of temperature lifts on the cycle performance and centrifugal compressor design. R1233zd(E) is used as the working fluid due to its excellent comprehensive performance for such applications [12]. Since R1233zd(E) is a dry fluid, a superheating of 8K is adopted accordingly as a safety measure to avoid wet compression under high temperature lifts [55]. It should be noted that working fluids have a strong impact on centrifugal compressor design. Further investigations could consider alternative candidates such as R1336mzz(Z) and CO₂ blends.

Table 2. Operating parameters of the HTHP system.

Parameters	Unit	Value
Heat source temperature	°C	80
Temperature drops of Heat source	K	20
Heat source heat flow rate	kW	1000
Pinch point temperature difference in heat exchangers	K	5
Subcooling at the condenser outlet	K	5
Superheating at the compressor inlet	K	5
Condensation temperature	°C	100~130

By using these cycle operating parameters, the design inputs of centrifugal compressors can be generated through the thermodynamic model, namely, $\dot{m}_f$, P_{eva} , T_1 and P_{cond} . Following that, these design inputs are passed to the centrifugal compressor design procedure to produce the optimal design and return the compressor efficiency to complete the calculation of the HTHP model. Thus, the compressor design is directly coupled to the cycle conceptual design of HTHPs, as illustrated by Fig. 4.

Table 3 summarizes the optimization range of design variables and constraints for the centrifugal compressor design. The corresponding values are based on the literature on centrifugal compressor design [47,53], which are evaluated to be appropriate to designing a physically feasible and efficient one. It should be noted that, as mentioned above, an upper limit constraint is imposed on $Ma_{1,rel}$ and $Ma_{1s,rel}$, respectively, to avoid infeasible designs in advance [36,49].

Table 3. Optimization range of design variables and constraints for the centrifugal compressor design.

Parameters	Description	Unit	Value
N_s	specific speed	–	0.4~0.8
ψ	tangential velocity coefficient	–	0.5~0.8
β_{2b}	impeller outlet blade angle	°	30~60
C_{1m}	inlet meridional velocity	m/s	60~120
κ	inlet hub-to-shroud diameter ratio	–	0.3~0.6
$Ma_{1,rel}$	relative Mach number at the impeller inlet	–	<0.9
$Ma_{1s,rel}$	relative tip Mach number at the impeller inlet	–	<1.4
t_{b1}	leading edge thickness	mm	2
t_{b2}	trailing edge thickness	mm	2
ϵ	clearance between the casing and the impeller	mm	5
ϵ_b	impeller back face clearance	mm	5
K_{SF}	skin friction coefficient	mm	0.05
$R_{D,diff}$	diffuser outlet to impeller outlet diameter ratio	–	1.65
$R_{b,diff}$	diffuser outlet to impeller outlet height ratio	–	1

For the comparative assessment of the resultant compressor schemes with the existing experience, different design characterization parameters are investigated. Table 4 lists the used characterization parameters [47,48].

Table 4. Optimization range of design variables and constraints for the centrifugal compressor design.

Parameters	Description	Unit	Definition
$R_{bD,out}$	impellor outlet height to diameter ratio	–	b_2/D_2
$R_{D,imp}$	impellor shroud diameter ratio	–	D_{1s}/D_2
$R_{DeHaler}$	DeHaler number	–	W_2/W_{1s}
$Ma_{2,abs}$	absolute Mach number at impeller outlet	–	$C_2/C_{2,sound}$
Ω_{flow1}	flow coefficient	–	C_{1m}/U_2
Ω_{flow2}	Outlet flow coefficient	–	C_{2m}/U_2
Ψ_{act}	actual tangential velocity coefficient	mm	C_{2u}/U_2
Φ_{work}	work coefficient	–	$\Delta h_{Euler}/U_2^2$

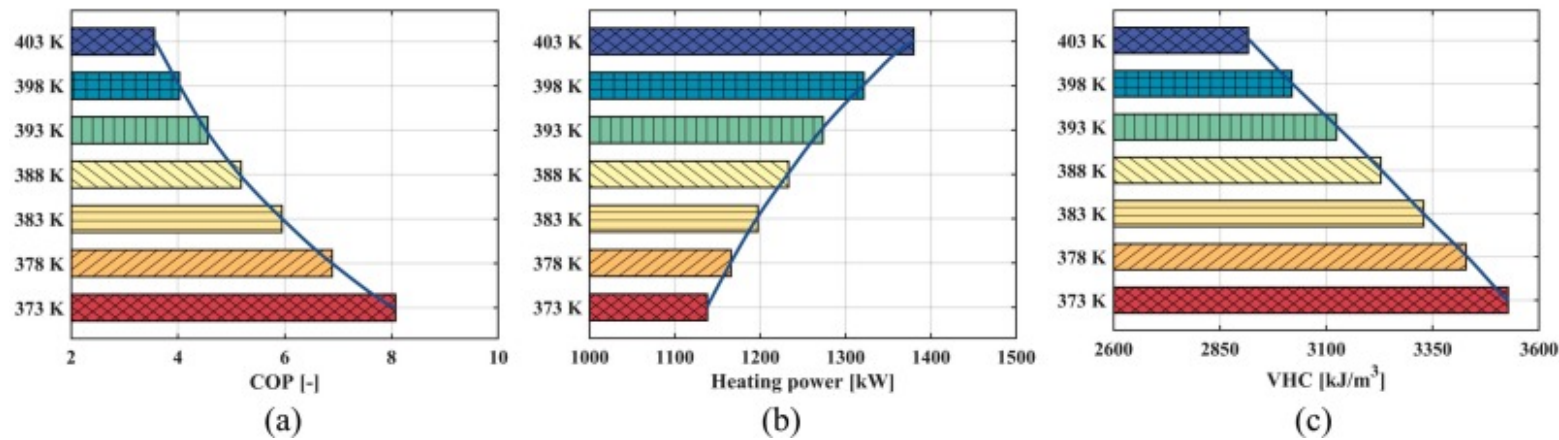
4. Results and discussions

The resultant optimal compressor designs at different condensation temperature are presented in Table 5. By incorporating the compressor design into the system model, a more realistic analysis on the HTHP can be obtained. The variations of different performance indicators with the condensation temperature (T_{cond}) are shown in Fig. 7. As expected, the COP decreases rapidly with T_{cond} increasing, accompanied by a downward trend in decrement. At the T_{cond} of 373K, the COP reaches 8.08, while reduces to 3.56, when T_{cond} moves to 403K, which corresponds to a temperature lift of 65K. This can be explained by the fact that the theoretical COP of HPs is inversely proportional to the temperature lift and directly proportional to T_{cond} . From Fig. 7(b), the heating power increases gradually with the increase of T_{cond} , influenced by the growth of the compression power. A relative increment of 21% is obtained as the T_{cond} changes from 373K to 403K. In terms of VHC, the results suggest that it descends linearly as T_{cond} rises. When T_{cond} varies from 373K to 403K, the VHC is reduced from 3527kJ/m³ to 2916kJ/m³, equivalent to a

relative decrease of 17%. The reason is that the condensation latent heat of the working fluid drops with T_{cond} rising. Since VHC is an indicator of compressor size, it implies that at the same heating load, the system size will inevitably increase as the supply temperature grows, resulting in a deterioration in economy.

Table 5. Optimal HTHP compressor designs under different condensation temperatures.

Parameters	Unit	373K	378K	383K	388K	393K	398K	403K
N_{des}	rpm	19,160	18,665	18,385	18,078	17,718	17,110	16,457
D_{1h}	mm	29	29	30	31	32	33	34
D_{1s}	mm	95	98	100	103	106	110	114
$\beta_{1b,h}$	°	19.49	19.65	19.31	19.50	19.70	19.91	19.67
$\beta_{1b,s}$	°	55.38	55.44	55.35	55.46	55.55	55.64	55.56
D_2	mm	161	176	188	204	215	233	251
b_2	mm	10	9	8	8	7	6	6
β_{2b}	°	39.83	39.59	39.90	46.71	43.38	44.92	46.09
$Z_{b,F}$	–	11	11	12	12	12	12	13
$Z_{b,S}$	–	11	11	12	12	12	12	13
L_a	mm	43	48	52	58	61	68	74
D_4	mm	266	290	310	337	355	384	415
b_4	mm	10	9	8	8	7	6	6

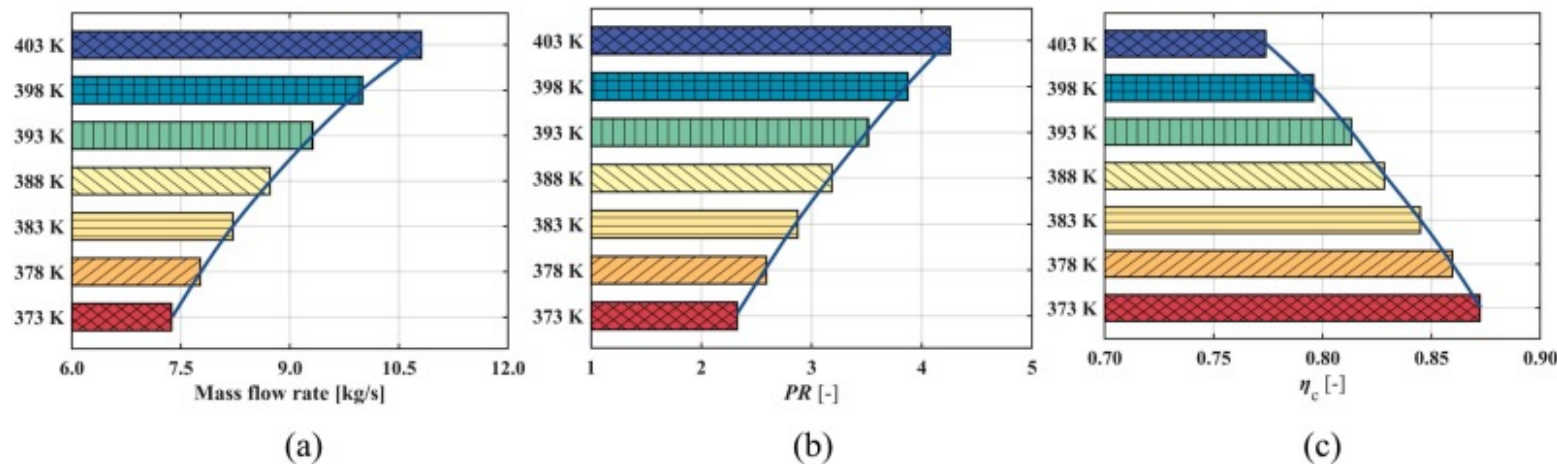


[Download: Download high-res image \(377KB\)](#)

[Download: Download full-size image](#)

Fig. 7. Effects of the condensation temperature on: (a) COP, (b) heating power, and (c) VHC.

Fig. 8 shows the effects of T_{cond} on the mass flow rate, pressure ratio, and isentropic efficiency of the compressor. As the T_{cond} increases, there is a growth in $\dot{m}_f$ with a similar trend to the heating power. This phenomenon results from the fact that with the T_{cond} rising, more heat is recovered by IHX, leading to a decline in the specific enthalpy at the evaporator. Additionally, the growth of $\dot{m}_f$ also means that a larger compressor is required at high temperature lifts. From Fig. 8(b), the PR changes from 2.33 to 4.26, when T_{cond} increases to 403 K, approximately proportional to the temperature lift. According to Fig. 8(c), the η_c ranges from 0.77 to 0.87 depending on the operating conditions, which are in alignment with the expectations for a well-designed centrifugal compressor [36]. Moreover, with T_{cond} rising, the η_c decreases gradually under the combined influence of $\dot{m}_f$ and PR . The main causes could be the increasing flow loss due the higher stage loads. This indicates that to ensure the high efficiency of centrifugal compressors, a multi-stage arrangement may be necessary, which in turn can result in a high investment cost. Hence, a compromise design of the HTHP system can be required to balance the efficiency and economy under high temperature lifts.

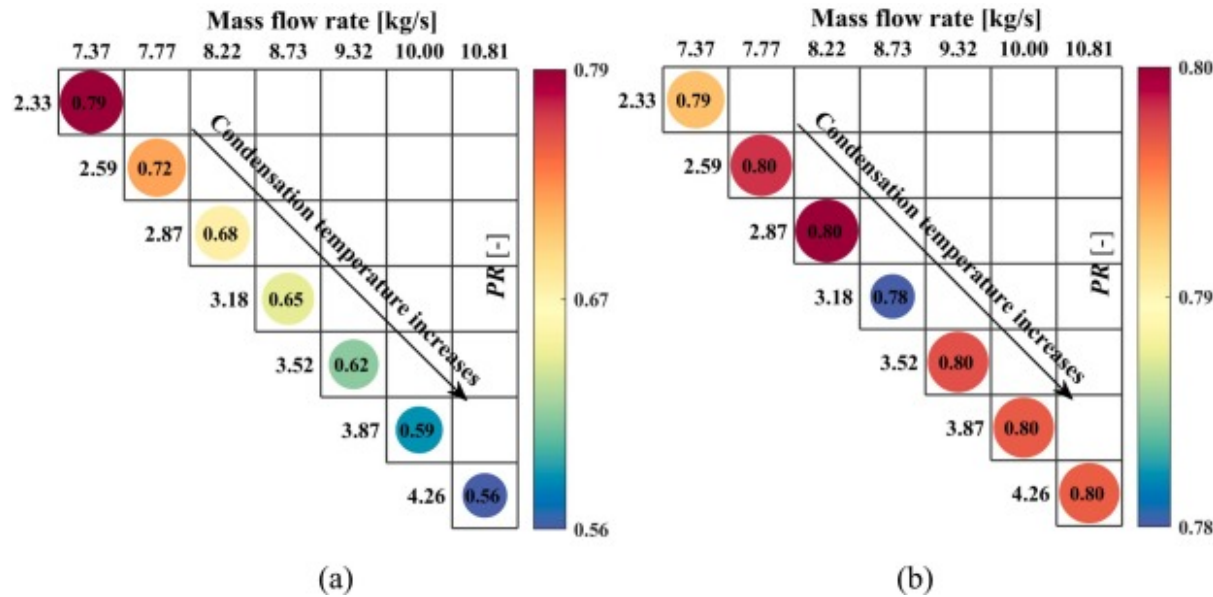


[Download: Download high-res image \(395KB\)](#)

[Download: Download full-size image](#)

Fig. 8. Effects of condensation temperature on: (a) mass flow rate, (b) PR, and (c) compressor efficiency.

Fig. 9 presents the optimal specific speeds and tangential velocity coefficients under different T_{cond} . The corresponding $\dot{m}_f$ and PR are also shown in the figure, which are the key design inputs of the compressor. As can be seen, there is a negative correlation between N_s and T_{cond} . When T_{cond} increases from 373 K to 403 K, the optimal N_s changes from 0.79 to 0.56. Since the physical meaning of N_s is the required rotational speed to obtain the unit energy head, it implies the main energy increase will rely on the large impeller outlet size. One possible explanation is the high N_s will correspond to a large relative speed at impeller inlet, thus more frictional losses. It should be noted that according to the well-developed Balje diagram of conventional centrifugal compressor, the optimal aerodynamic efficiency can be achieved when N_s is between 0.4~0.8 [53]. This suggests that the resultant designs conform to the criterion. Nevertheless, the results also indicate the suboptimal design can be obtained with a fixed N_s . From Fig. 9(b), the optimal ψ are near the upper bound 0.8. However, relatively noticeable deviations still exist under certain T_{cond} due to the strong nonlinearity of the model itself.



[Download: Download high-res image \(244KB\)](#)

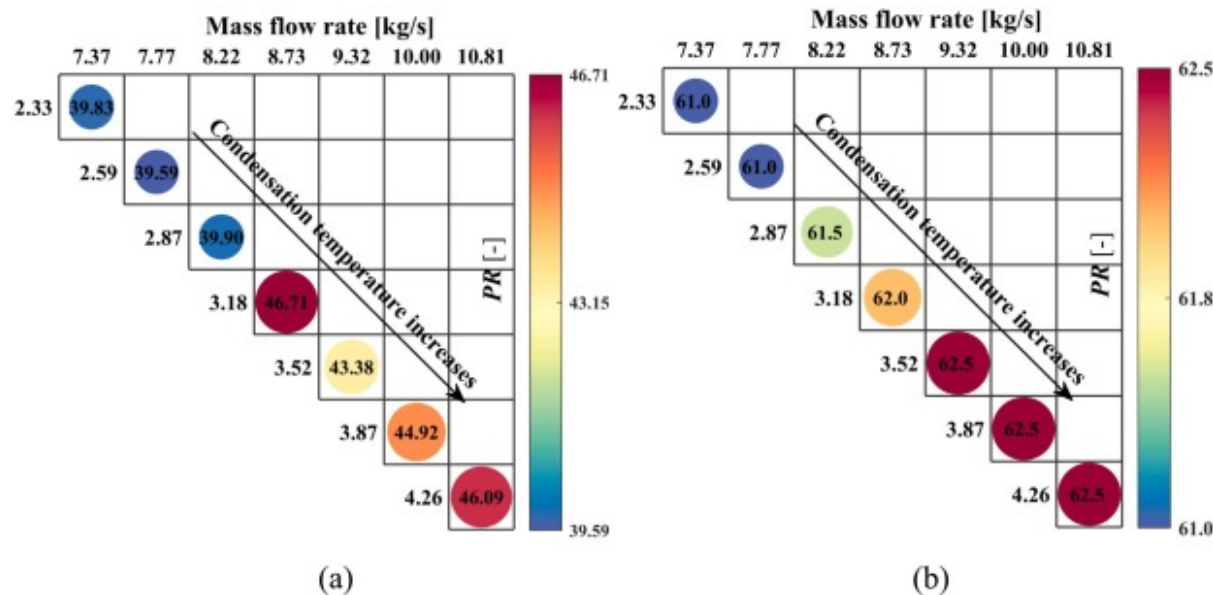
[Download: Download full-size image](#)

Fig. 9. Optimal specific speeds (a) and tangential velocity coefficients (b) under different T_{cond} .

Fig. 10 displays the variations of the optimal impeller outlet blade angle and inlet meridional velocity with T_{cond} . As shown by Fig. 10(a), the optimal β_{2b} varies to a certain extent, affected by the increase of $\dot{m}_f$ and PR . Nevertheless, the optimal ones are within $39^\circ \sim 47^\circ$. Additionally, the β_{2b} is about 39.5° at small $\dot{m}_f$ and PR , while scatters around about 45° at large $\dot{m}_f$ and PR .

Theoretically, β_{2b} has a great influence on the stage pressure ratio. For the backward swept impeller, the larger β_{2b} , the smaller the added energy, and consequently, a reduced PR . Meanwhile, the growth of β_{2b} will cause the increase of the degree of reaction and stage efficiency. Therefore, some researchers suggest that a suitable range of β_{2b} is $0.3 \sim 0.6$ in the conventional compressor design [48]. Again, the resultant HTHP centrifugal compressors also exhibit the consistency in β_{2b} , with a more concentrated pattern for the investigated case. Fig. 10(b) shows that optimal C_{1m} should be very close to the lower bound in order to minimize the $Ma_{1s,rel}$. With a slight increasing tendency, the C_{1m} varies from 61 m/s to 62.5 m/s as at the T_{cond} increases to 403 K. As for the optimal κ , it always keeps at 0.3, namely, the lower bound according to optimization results. It's worth noting that one major difference of HTHP centrifugal compressor design from the conventional one is that the working fluids, such as R1233zd(E),

typically have a low sound speed, far lower than air, which can easily cause the design issues of transonic flow. This highlights the necessity of the specialized impeller design.



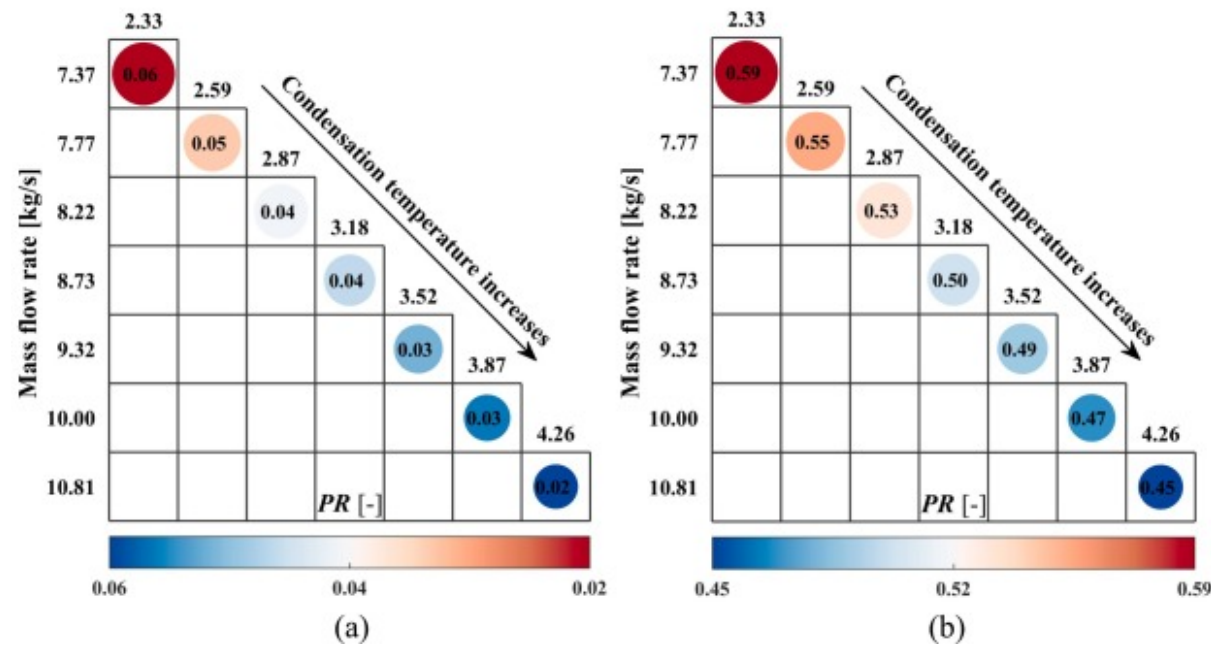
[Download: Download high-res image \(243KB\)](#)

[Download: Download full-size image](#)

Fig. 10. Optimal impeller outlet blade angles (a) and inlet meridional velocities (b) under different T_{cond} .

Fig. 11 shows the impeller outlet height to diameter ratios and impeller shroud diameter ratios of the optimal designs. A clear pattern of the $R_{bD,out}$ is that with T_{cond} rising, it falls gradually. This is logic, since the PR is raised by the increasing T_{cond} , resulting in a larger fluid density at the impeller outlet. Besides, the impeller diameter also tends to increase so as to reach a larger PR . At the low T_{cond} , the value of $R_{bD,out}$ is about 0.06, and it reduces to 0.02 when T_{cond} is raised to 403K. In fact, the $R_{bD,out}$ is directly connected with the outlet relative velocity W_2 . For a given D_2 , the growth of $R_{bD,out}$ will reduce W_2 , which increases the impeller diffusion. Since the efficiency of the impeller is usually higher than that of the diffuser, an appropriate increase in $R_{bD,out}$ can improve the compressor performance. However, the increase in $R_{bD,out}$ also brings a larger W_1/W_2 , leading to a premature flow separation and increased mixing losses. On this basis, an appropriate $R_{bD,out}$ usually is generally considered to lie between 0.02 and 0.08, depending on the operating conditions. Thus, the optimal impeller designs are also compliant with the standard. Fig. 11(b) suggests that the larger the T_{cond} , the smaller the $R_{D,imp}$, affected by the coordinated variation of $\dot{m}_f$ and PR . The value of

$R_{D,imp}$ varies from 0.45~0.59 with the increase of T_{cond} . Since $R_{bD,out}$ directly impacts the flow within the impeller, values that are either excessively high or low will be detrimental to the performance. The existing design experience of the impeller suggests that $R_{bD,out}$ should be 0.45~0.60 to achieve a desirable performance. The resultant designs meet the requirement very well.



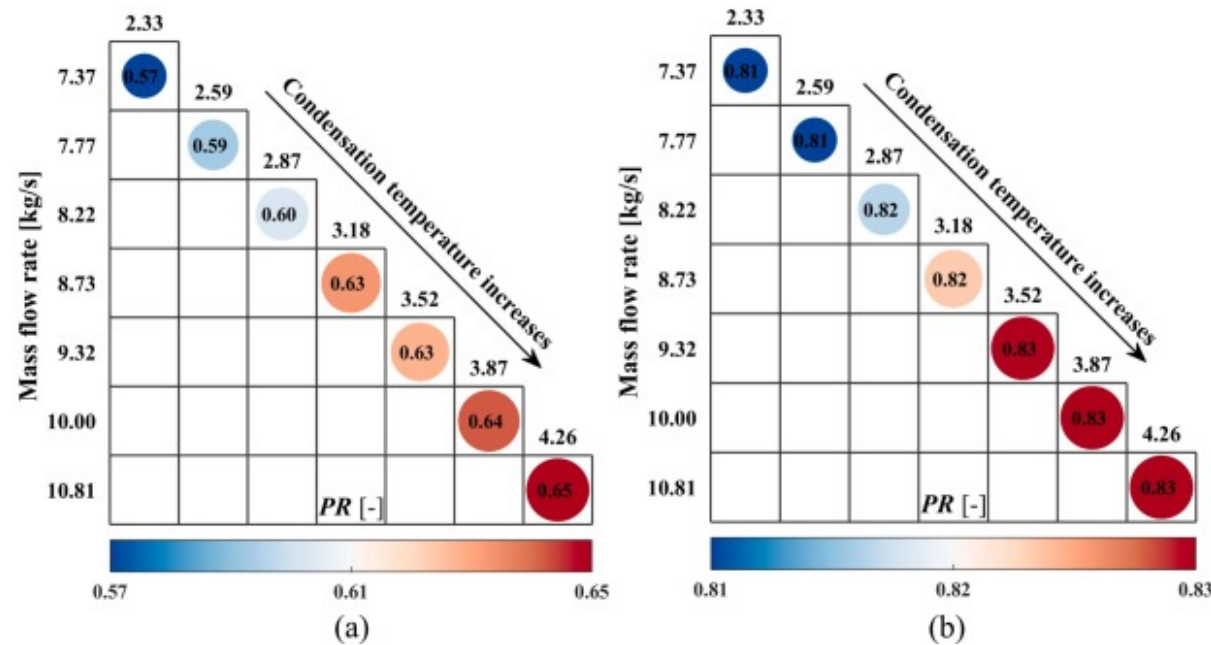
[Download: Download high-res image \(249KB\)](#)

[Download: Download full-size image](#)

Fig. 11. Impellor outlet height (a) to diameter ratios and impellor shroud diameter ratios (b) of the optimal designs.

Fig. 12 gives the DeHaler numbers and inlet relative tip Mach numbers of the optimal compressor designs. As shown in Fig. 12(a), with the rise of T_{cond} , $R_{DeHaler}$ goes up. At a temperature lift of 35K, the value is 0.57, while it increases to 0.65 when the temperature lift changes to 65K. Since $R_{DeHaler}$ represents the ratio of W_2 to W_{1s} , this parameter can reflect the flow losses in the blade channels. It has been reported that a satisfactory range of $R_{DeHaler}$ should be 0.55~0.70 [48], which is equivalent to maintain a moderate pressure rise in the impeller. As demonstrated by the results, the optimal designs are in good agreement in this aspect. According to Fig. 12(b), the $Ma_{1s,rel}$ show a growth trend as T_{cond} increases. When T_{cond} changes from 373K to 403K, $Ma_{1s,rel}$ increases from 0.81 to 0.83. In general, the impellers with high $Ma_{1s,rel}$ have a relatively poor overload capacity and

significant flow loss. For this reason, it is usually recommended that a maximum allowable $Ma_{1s,rel}$ should be 0.85~0.92 to attain a successful design. This indicates that the compressors designed should have a good overall performance.



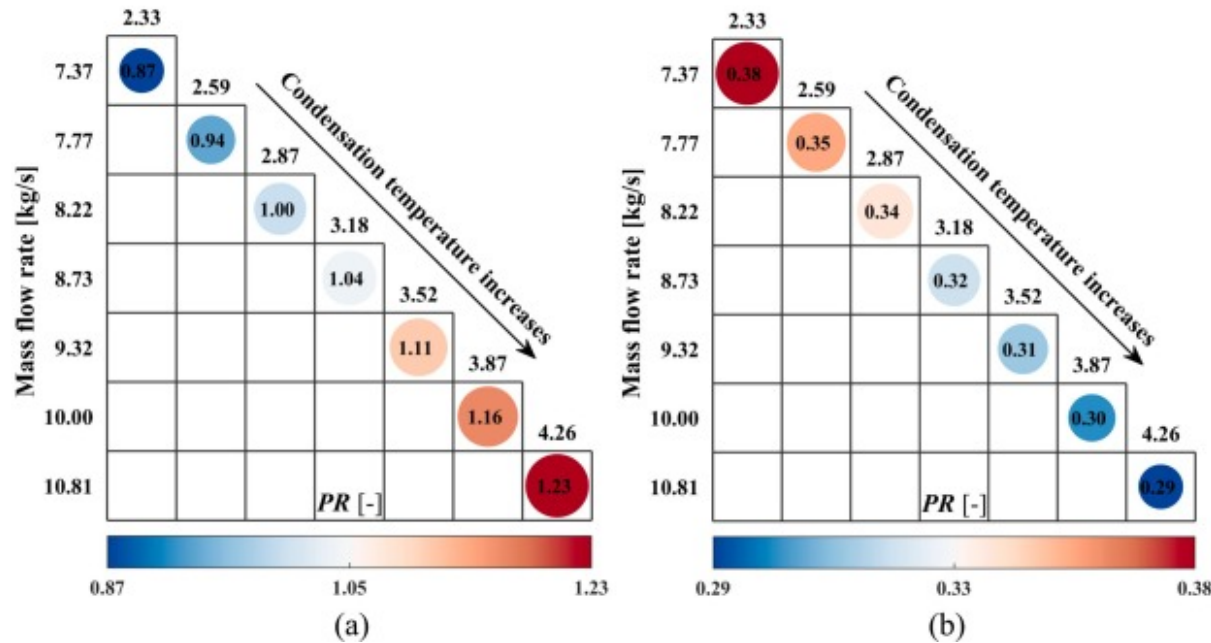
[Download: Download high-res image \(254KB\)](#)

[Download: Download full-size image](#)

Fig. 12. DeHaler numbers (a) and inlet relative tip Mach numbers (b) of the optimal designs.

Fig. 13 shows the outlet absolute Mach numbers and flow coefficients of the optimal designs. The $Ma_{2,abs}$ has a clear upward trend as T_{cond} rises, influenced by the variations of $\dot{m}_f$ and PR . At the T_{cond} of 373K, $Ma_{2,abs}$ is 0.87, while it increases to 1.23 with T_{cond} rising to 403K. Since $Ma_{2,abs}$ is associated with the losses in the following components, it implies at high PR , stator losses will experience an inevitable rapid increase. It should be stressed that $Ma_{2,abs}$ heavily relies on fluid properties. The working fluids with heavy mole weight incline to have a low sound speed, thus a high $Ma_{2,abs}$. This can explain the rapid increase of $Ma_{2,abs}$ for the investigated case. Also, it indicates that a focus should be placed on the advanced diffuser design. As for Ω_{flow1} , it declines with the increase of T_{cond} . As T_{cond} increases to 403K, it falls from 0.38 to 0.29. Previous research has shown that Ω_{flow1}

correlates with the outlet flow angle and thus the diffuser efficiency. Moreover, an inappropriate Ω_{flow1} can cause flow separation. Accordingly, it is recommended to have Ω_{flow1} between 0.25~0.40. Again, the results well meet the guideline.



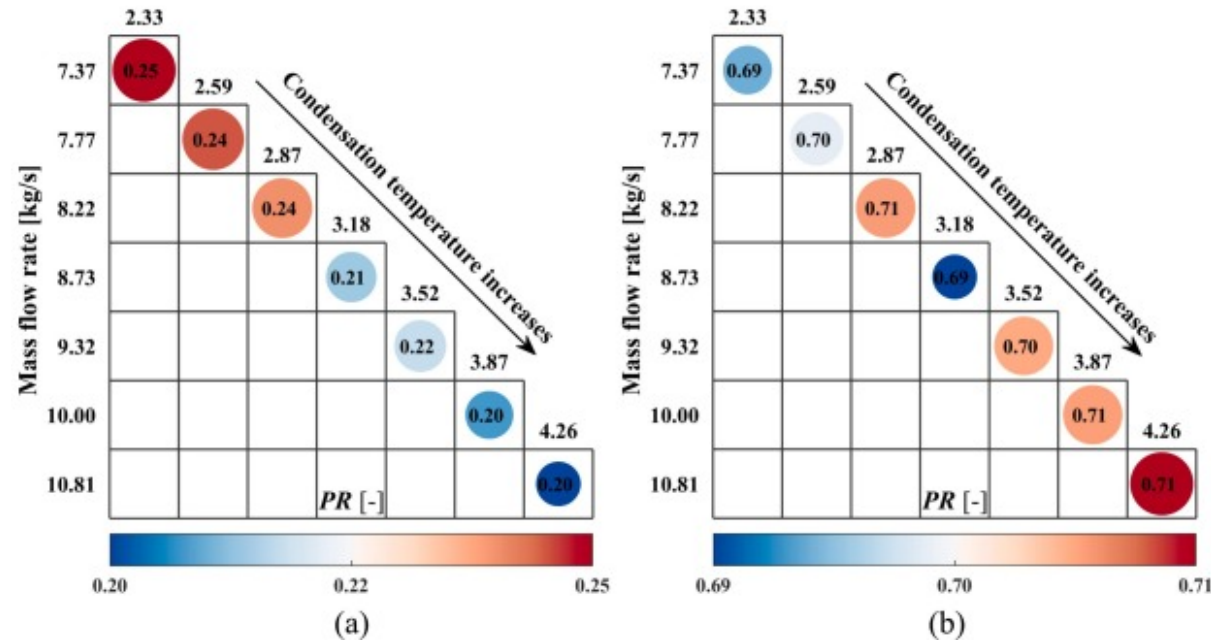
[Download: Download high-res image \(251KB\)](#)

[Download: Download full-size image](#)

Fig. 13. Outlet absolute Mach numbers (a) and flow coefficients (b) of the optimal designs.

Fig. 14 presents the outlet flow coefficients and actual tangential velocity coefficients of the optimal designs. Like Ω_{flow1} , Ω_{flow2} also exhibits a decreasing trend with T_{cond} increasing, while it has a smaller value comparatively. When T_{cond} varies from 373 K to 403 K, Ω_{flow2} drops from 0.25 to 0.20. The reason lies in that there is a negative correlation between Ω_{flow2} and the impeller work, which means to obtain a high PR, a small Ω_{flow2} is preferable. However, if Ω_{flow2} is too small, the average flow velocity in the impeller passage will inevitably decrease. Thereby, the stage performance can deteriorate significantly influenced by the axial vortex. This hints at a decrease in compressor performance at high PR. From the existing design experience, the value of Ω_{flow2} usually ranges from 0.11 to 0.31 depending on β_{2b} . The results also show good agreement. Fig. 13(b) shows that Ψ_{act} fluctuates

around 0.70 slightly with the change of T_{cond} . Moreover, Ψ_{act} is smaller compared with Ψ , resulting from the phenomenon of velocity slip.

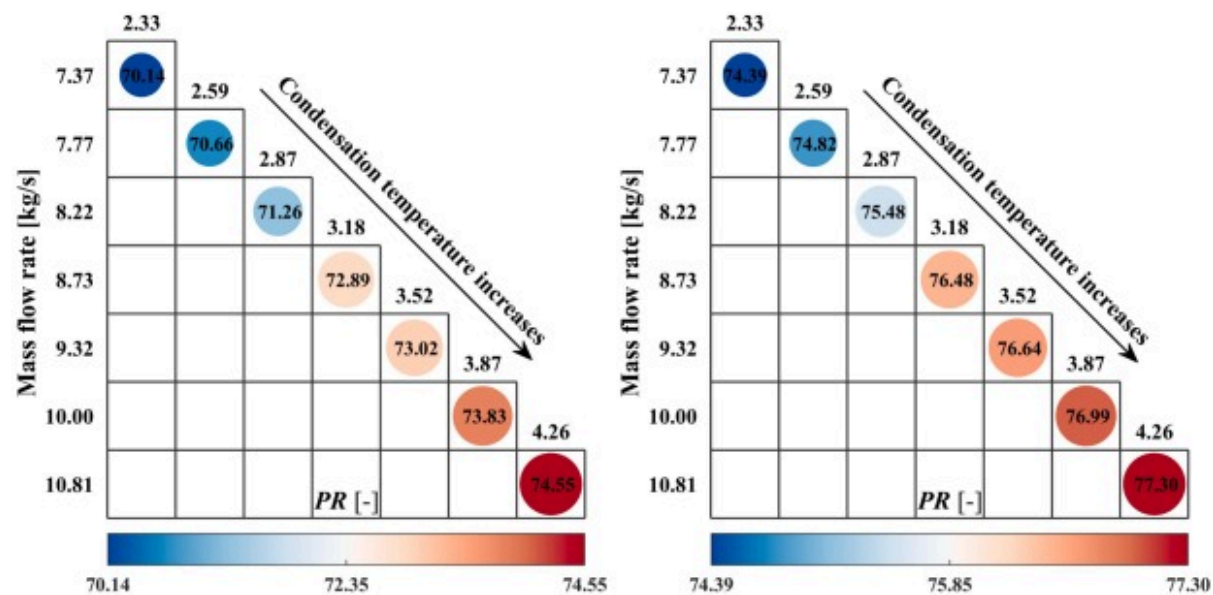


[Download: Download high-res image \(251KB\)](#)

[Download: Download full-size image](#)

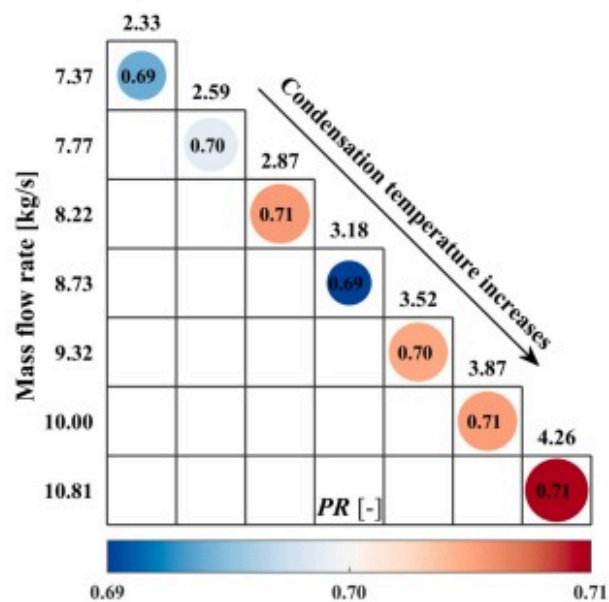
Fig. 14. Outlet flow coefficients (a) and actual tangential velocity coefficients (b) of the optimal designs.

Fig. 15 presents the absolute flow angles at the impeller outlet and diffuser outlet and work coefficients of the optimal designs. As shown in Fig. 15(a), α_2 increases from 70.14° to 74.55° with T_{cond} rising. In fact, α_2 is influenced by the $R_{\text{bD,out}}$. Normally, the smaller the $R_{\text{bD,out}}$, the larger the α_2 . Research suggests that decreasing α_2 can help improve the operating performance of the impeller and vaneless diffuser, and increase the stability of the flow in the diffuser. On this basis, it is recommended to have an α_2 smaller than $76^\circ \sim 78^\circ$. Thus, the resultant optimal designs are well in line with the design criterion. As to α_4 , it also exhibits an increasing trend with the rise of T_{cond} , with its value changing from 74.39° to 77.33° . Moreover, the α_4 is larger than α_2 due to friction. According to Fig. 15(c), the Φ_{work} is between 0.69 and 0.71. The value jumps around 0.70, with a deviation of about 0.01. In general, this suggests a good stage performance.



(a)

(b)



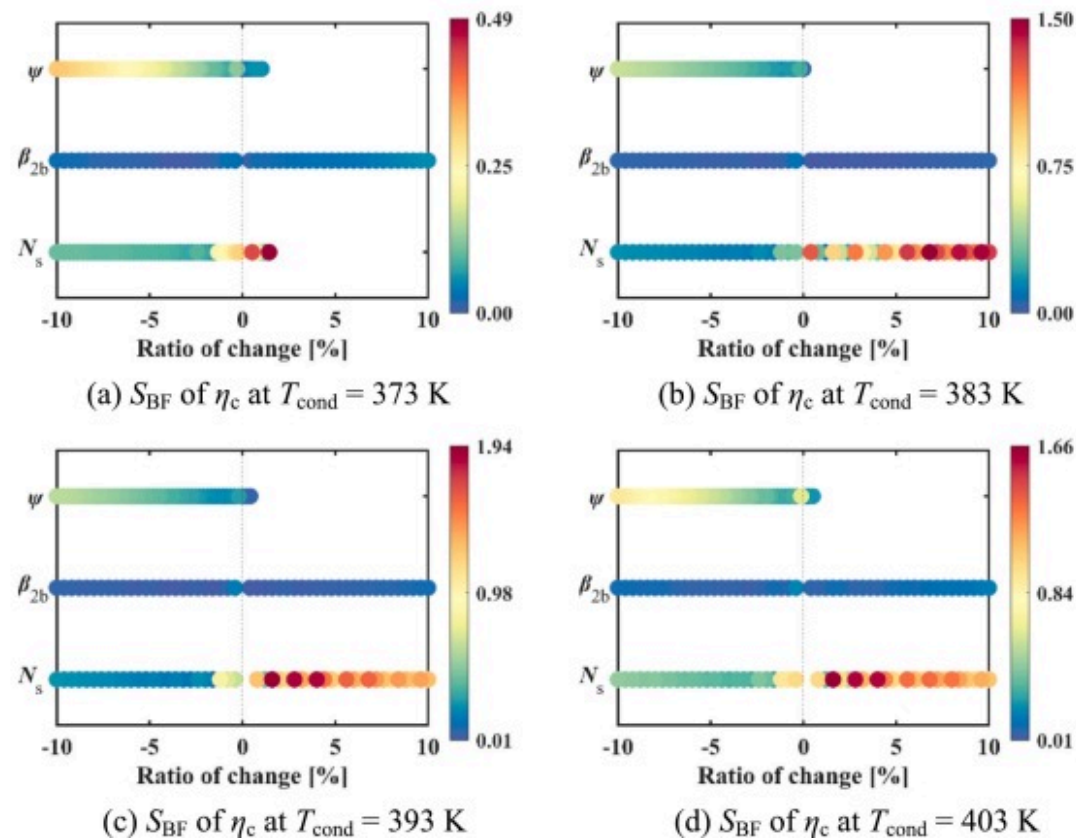
(c)

[Download: Download high-res image \(396KB\)](#)

[Download: Download full-size image](#)

Fig. 15. Absolute flow angles at the impeller outlet (a) and diffuser outlet (b) and work coefficients (c) of the optimal designs.

Fig. 16 shows the sensitivity of η_c to the design variables N_s , β_{2b} , and ψ at different T_{cond} . The change of the design variables is assumed to be $\pm 10\%$ of the optimal value, constrained by the optimization ranges simultaneously. It can be found that there is a clear nonlinearity for η_c in general as the design variables change. Among the design variables, η_c has the lowest sensitivity to β_{2b} at various T_{cond} , and the S_{BF} of β_{2b} remains relatively stable with the change of β_{2b} . A trend for ψ is that the S_{BF} gradually increases as ψ decreases for all the cases. As to N_s , it has a relatively small impact on η_c when it is reduced, whereas a significant but unstable effect on the S_{BF} is observed when it is increased. This suggests that an inappropriate choice of N_s can result in a poor compressor performance when relying on an empirical design strategy. It also can be found that T_{cond} is related to the sensitivity of η_c to the design variables. As T_{cond} increases, the sensitivity of η_c varies markedly. This highlights the necessity of a specialized design based on the specific working conditions of HTHPs.

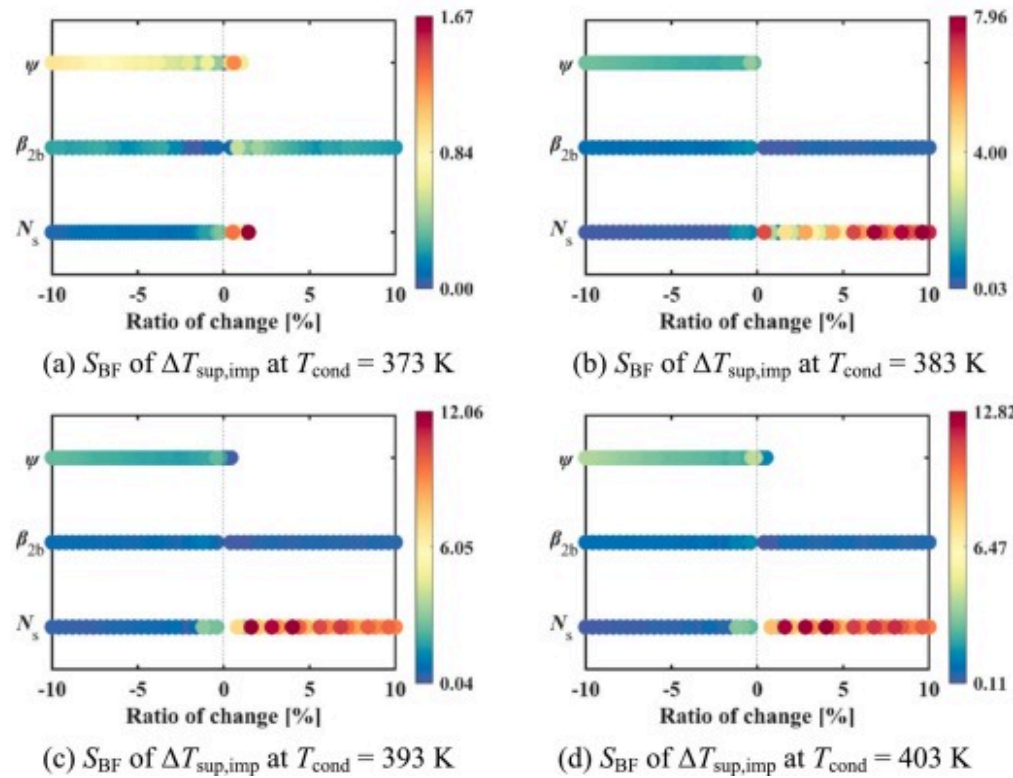


[Download: Download high-res image \(284KB\)](#)

[Download: Download full-size image](#)

Fig. 16. Sensitivity of η_c to design variables at different T_{cond} .

Fig. 17 displays the sensitivity of the superheat at the impeller outlet ($\Delta T_{sup,imp}$) to the design variables N_s , β_{2b} , and ψ at different T_{cond} . This indicator can quantitatively reflect the risk of wet compression. As shown in the figure, N_s exhibits a trivial impact on $\Delta T_{sup,imp}$ when it is decreased, while it has a strong and highly nonlinear effect on $\Delta T_{sup,imp}$ when it is increased. The S_{BF} of β_{2b} at different T_{cond} indicates that $\Delta T_{sup,imp}$ has a minimal sensitivity to β_{2b} . As for ψ , the results show that it generally has a moderate impact on $\Delta T_{sup,imp}$. Additionally, there is a clear impact on $\Delta T_{sup,imp}$ with T_{cond} changing. In contrast with Fig. 16, it can be found that a similarity exists in the effect of the design variables on η_c and $\Delta T_{sup,imp}$. This implies that the fluid condition at the impeller outlet has a strong connection with the overall performance of the compressor.



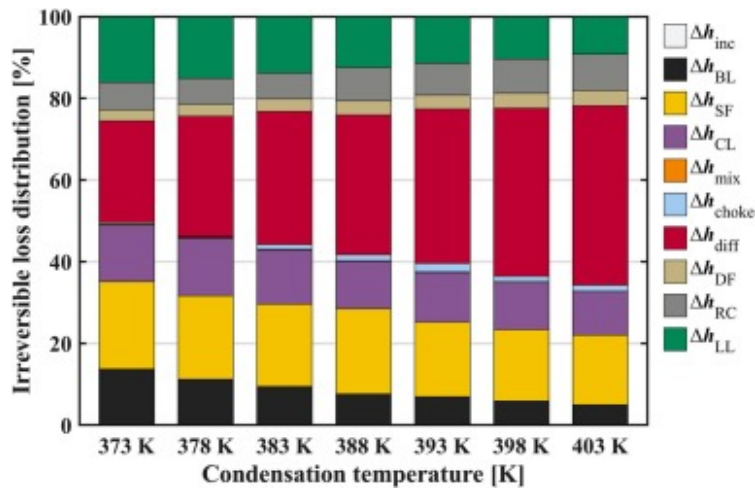
[Download: Download high-res image \(275KB\)](#)

[Download: Download full-size image](#)

Fig. 17. Sensitivity of the impeller outlet superheat to design variables at different T_{cond} .

Fig. 18 displays the irreversible loss distributions of the optimal designs at different T_{cond} . It shows that the main causes of irreversible losses are the diffuser loss, skin friction loss, leakage loss, clearance loss, recirculation loss, and blade loading loss for all the operating conditions. Specifically, Δh_{diff} accounts for the largest proportion, followed by Δh_{SF} . As T_{cond} rising, Δh_{diff} shows a significant increasing trend, from 25% at the T_{cond} of 373K to 44% at the T_{cond} of 403K. This is attributed to the fact that there is a rapid growth in the diffuser size with the increase of PR . Meanwhile, α_2 becomes larger under the combined effect of $\dot{m}_f$ and PR . Thus, the flow path in the diffuser grows rapidly, leading to the marked increase in friction loss. It indicates that pressure rise mainly occurs in the diffuser under a large temperature lift, and thus a reasonable diffuser design scheme is crucial. As to Δh_{SF} , its proportion gradually declines from 21% to 17% with increasing T_{cond} . Another notable loss is Δh_{LL} , which decreases from 16%

to 9% with the rise of T_{cond} . The same trend can also be seen in Δh_{BL} and Δh_{CL} . By contrast, the proportion of Δh_{RC} grows as T_{cond} increases, with the value changing from 7% to 9%. This is logical since Δh_{RC} originates from the recirculation back into the impeller section after leaving the impeller outlet, driven by the high pressure at the impeller outlet.

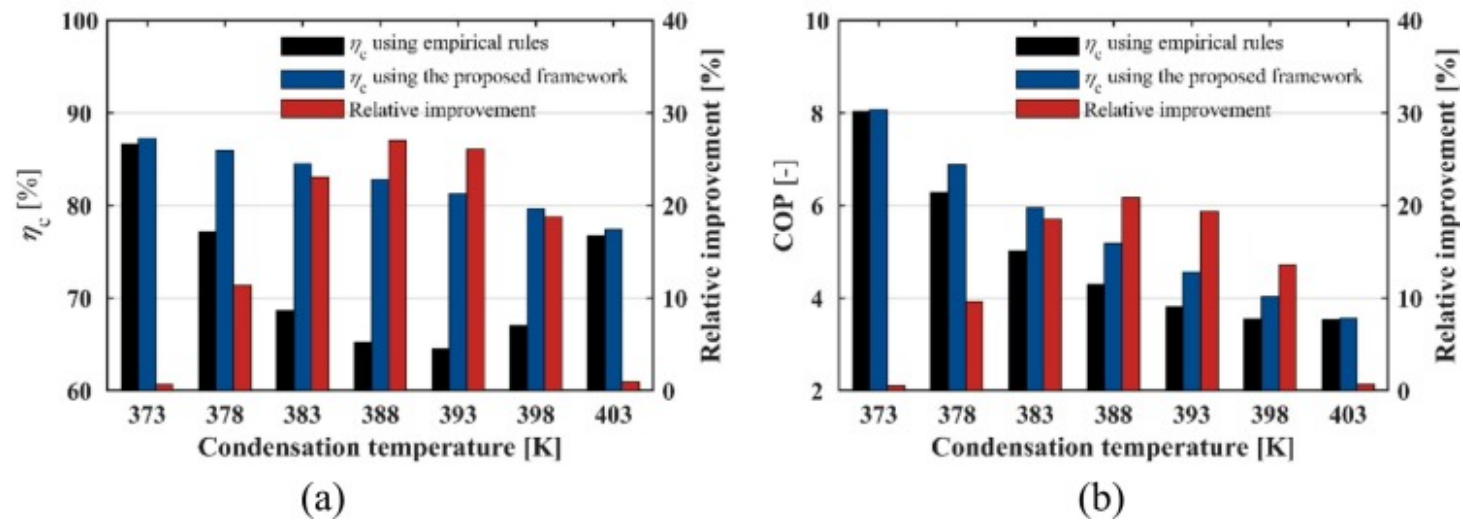


[Download: Download high-res image \(149KB\)](#)

[Download: Download full-size image](#)

Fig. 18. Irreversible loss distributions of the optimal designs under different T_{cond} .

To evaluate the superiority of the proposed optimization framework, a comparison with the empirical design strategy is conducted, for which the specific speed is fixed at 0.8, as adopted in Ref. [23]. The corresponding comparison results are shown in Fig. 19. It can be seen that the proposed optimization framework achieves better η_c and COP across all T_{cond} conditions, compared with the empirical design strategy. With the increase of T_{cond} , the relative improvement of η_c and COP increase first and decrease later. The maximal relative improvements are 27% and 21% for η_c and COP, respectively. Obviously, the increase in COP is mainly due to the improvement in η_c . The main reason for the variation is that there are clear differences in the sensitivity of η_c to N_s under different operating conditions, as shown in Fig. 16.



[Download: Download high-res image \(245KB\)](#)

[Download: Download full-size image](#)

Fig. 19. Comparisons of η_c (a) and COP (b) with the design approach using the empirical rules.

5. Conclusions

In this work, a holistic design framework for HTHP centrifugal compressors is proposed, with wet compression prevention and global optimization considered. The real fluid thermophysical properties are used to address the flow non-ideality in HPHHP applications. By adopting generic design parameters, the design framework enables the effective utilization of existing domain knowledge. A strategy based on the identification of stage component fluid conditions is developed to avoid possible wet compression at large temperature lifts. Moreover, a global optimization algorithm coupled with cycle conceptual design is incorporated to obtain the optimal design.

In addition, a case study of the HTHP with centrifugal compressors is performed based on the proposed design framework. The correlations of cycle operating parameters, cycle performance, and compressor design are analyzed. A comprehensive comparative assessment of resultant design schemes is carried out with key design characterization parameters. Furthermore, the corresponding irreversible loss distributions are also investigated. The main findings are summarized as follows:

- (1) The performance of centrifugal compressors exhibits clear dependence on the temperature lift. With T_{cond} rising from 373K to 403K, the optimal η_c declines from 0.87 to 0.77. Accordingly, the COP and VHC drop from 8.08 to 3.56 and from 3527 kJ/m³ to 2916 kJ/m³, respectively.
- (2) To achieve the desired pressure ratio under large temperature lift, the optimal design inclines to have a larger impeller outlet size. To minimize the $Ma_{1s,\text{rel}}$, the optimal C_{1m} needs to be very close to the lower bound, with the value ranging from 61 m/s to 62.5 m/s, and the κ should be as small as possible.
- (3) The design characterization parameters of the optimal designs show good agreement with the existing design experience. However, influenced by the low sound speed characteristic of the working fluid, the $Ma_{2,\text{abs}}$ shows a significant increase as T_{cond} grows. When T_{cond} changes from 373K to 403K, the variation in $Ma_{2,\text{abs}}$ corresponds from 0.87 to 1.23.
- (4) The main causes of irreversible losses in the compressor include Δh_{diff} , Δh_{SF} , Δh_{LL} , Δh_{BL} , Δh_{CL} , and Δh_{RC} . Particularly, Δh_{diff} accounts for the largest proportion, followed by Δh_{SF} . A clear pattern is that Δh_{diff} experienced a rapid growth with the temperature lift increase, highlighting the importance of the further improvement of the diffuser.

This study mainly focuses on developing a holistic mean-line based design framework of HTHP centrifugal compressors considering wet compression prevention and global optimization. A broader validation under high-temperature extreme-condition applications remains to be conducted limited by the absence of public data in this regard. In the future, authors would like to design and experimentally investigate HTHP centrifugal compressors with potential working fluids to systematically discuss the proposed methodology and provide meaningful references for future research in this field. Moreover, the practical applicability of the proposed methodology to ultra-high-temperature HTHP scenarios will be explored.

CRedit authorship contribution statement

Yinlian Yan: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fubin Yang:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Hongguang Zhang:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Nanqiao Wang:** Writing – review & editing,

Methodology. **Yonghong Xu:** Writing – review & editing, Funding acquisition. **Jiabao Cheng:** Methodology. **Zhong Ge:** Writing – review & editing. **Jian Li:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is sponsored by the National Natural Science Foundation of China (No. [52476001](#)) and Beijing Natural Science Foundation (Grant No. [3244039](#)). The authors sincerely thank the reviewers for their valuable and insightful comments on this research.

Appendix A. Calculation details of impeller internal and parasitic losses

The incidence loss occurs when there is a misalignment between the inflow and the inlet blade angle, which is described by [\[56\]](#):

$$\Delta h_{\text{inc}} = W_1^2 \sin^2 (\beta_{1,\text{opt}} - \beta_1) / 2 \quad (\text{A1})$$

$$\beta_{1,\text{opt}} = \tan^{-1} ((A_1 / A_{1,\text{th}}) \tan (\beta_{1\text{b}})) \quad (\text{A2})$$

where $\beta_{1,\text{opt}}$ is the blade angle at which no misalignment exist. It is obvious that the corresponding blade angle should match perfectly with the flow under the design conditions, which means the incidence losses are negligible for the design aim.

The blade loading loss is associated with the momentum loss due to the boundary layers on the blade surfaces, the correlation developed by Coppage et al. [\[57\]](#) is considered:

$$\Delta h_{\text{BL}} = 0.05 D_f^2 U_2^2 \quad (\text{A3})$$

$$D_f = 1 - W_2/W_1 + K_d \Delta h_{\text{Euler}} W_2 / (U_2^2 W_{1s}) (Z_{b,\text{eq}} (1 - D_{1s}/D_2) / \pi + 2D_{1s}/D_2)^{-1} \quad (\text{A4})$$

$$Z_{b,\text{eq}} = Z_{b,\text{F}} + Z_{b,\text{S}} L_S / L_F \quad (\text{A5})$$

$$K_d = \begin{cases} 0.75 & (\text{without splitters}) \\ 0.6 & (\text{with splitters}) \end{cases} \quad (\text{A6})$$

where D_f is the diffusion factor which measures the total deceleration of the flow.

The skin friction loss accounts for the wall friction losses of the flow in the impeller passage and can be predicted by [58]:

$$\Delta h_{\text{SF}} = 2K_{\text{SF}} L_{\text{hyd}} W_{\text{ave}}^2 / D_{\text{hyd}} \quad (\text{A7})$$

$$W_{\text{ave}} = (W_{1h} + W_{1s} + 2W_2) / 4 \quad (\text{A8})$$

$$L_m = \pi (D_2 - D_{1h} + (D_{1h} - D_{1s}) / 2 - b_2 + 2L_a) / 8 \quad (\text{A9})$$

$$L_{\text{hyd}} = L_m / \cos((\beta_1 + \beta_{2b}) / 2) \quad (\text{A10})$$

$$D_{\text{hyd}} = \pi D_{1M} \cos(\beta_1) / Z_{b1} b_1 / (\pi D_{1M} \cos(\beta_1) / Z_{b1} + b_1) + \pi D_2 \cos(\beta_{2b}) / Z_b b_2 / (\pi D_2 \cos(\beta_{2b}) / Z_b + b_2) \quad (\text{A11})$$

$$K_{\text{SF}} = \begin{cases} K_{\text{SF},\text{ts}} & \text{Re}_e < 60 \\ K_{\text{SF},\text{ts}} + (K_{\text{SF},\text{tr}} - K_{\text{SF},\text{ts}}) (1 - 60/\text{Re}_e) & \text{Re}_e \geq 60 \end{cases} \quad (\text{A12})$$

$$K_{\text{SF},\text{ts}}^{-1/2} = -4 \log_{10} (1.255/\text{Re}/K_{\text{SF},\text{ts}}^{1/2}) \quad (\text{A13})$$

$$K_{\text{SF},\text{tr}}^{-1/2} = -4 \log_{10} (k_s/3.71/D_{\text{hyd}}) \quad (\text{A14})$$

$$\text{Re}_e = (\text{Re} - 2000) k_s / D_{\text{hyd}} \quad (\text{A15})$$

where K_{SF} is the skin friction coefficient, W_{ave} is the average relative velocity through the impeller passage, L_{hyd} is the hydraulic length, and D_{hyd} is the hydraulic diameter.

The clearance loss is produced by the flow from the high-pressure side to the low-pressure side of the blade through the tip clearance, which can be measured by [58]:

$$\Delta h_{CL} = 0.6\epsilon / b_2 C_{2u} (\pi (D_{1s}^2 - D_{1h}^2) C_{2u} C_{1m} / (0.5 (D_2 - D_{1s}) (1 + \rho_2 / \rho_1) Z_b b_2)^{1/2} \quad (A16)$$

where ϵ is the clearance between the casing and the impeller.

The mixing loss originates from the finite thickness of the blade which leads to a non-uniform flow at the impeller outlet. To calculate the mixing losses, the correlation from [59] is used:

$$\Delta h_{mix} = 0.5 (C_{2m,wake} - C_{2m,mix})^2 \quad (A17)$$

$$D_{eq} = 0.5 (W_1 + W_2 + 2\pi D_2 \Delta h_{Euler} / U_2 / Z_{b,eff} / L_{hyd}) / W_2 \quad (A18)$$

$$W_{sep} = \begin{cases} W_2 & D_{eq} \leq 2 \\ W_2 D_{eq} / 2 & D_{eq} > 2 \end{cases} \quad (A19)$$

$$C_{2m,mix} = C_{2m} (1 - Z_b * t_2 / \pi / D_2) \quad (A20)$$

$$C_{2m,wake} = (W_{sep}^2 - W_{2u}^2)^{1/2} \quad (A21)$$

where $C_{2m,mix}$ and $C_{2m,wake}$ are the meridional velocities when the wake mixing occurs.

The choke loss results from the supersonic flow at impeller throat section. To evaluate the choke losses, the following correlation is adopted [60]:

$$\Delta h_{choke} = \begin{cases} 0 & X \leq 0 \\ 0.25 W_1^2 (0.05 X + X^7) & X > 0 \end{cases} \quad (A22)$$

$$X = m_f / m_{f,choke} - (11 - 10 m_f / m_{f,choke})^4 \quad (A23)$$

where $m_{f,choke}$ is the mass fluid rate at which the choking occurs.

The disc friction loss is produced by the shearing action between the disk and the fluid. The losses can be predicted by [61]:

$$\Delta h_{\text{DF}} = K_{\text{DF}} (\rho_1 + \rho_2) D_2^2 U_2^3 / m_f / 32 \quad (\text{A24})$$

$$K_{\text{DF}} = \begin{cases} 3.7(\epsilon_b/b_2)^{0.1} / \text{Re}_2^{0.5} & \text{Re}_2 \leq 3 \cdot 10^5 \\ 0.102(\epsilon_b/b_2)^{0.1} / \text{Re}_2^{0.2} & \text{Re}_2 > 3 \cdot 10^5 \end{cases} \quad (\text{A25})$$

where K_{SF} is the disc friction factor.

The recirculation loss is caused by the impeller outflow recirculating back into the impeller, which can be estimated by [62]:

$$\Delta h_{\text{RC}} = 8 \cdot 10^{-5} \sinh(3.5\alpha_2^3) D_f^2 U_2^2 \quad (\text{A26})$$

The leakage loss results from the fact that part of the flow gets leaked through seals. The correlation from [59] is employed to measure the losses:

$$\Delta h_{\text{LL}} = m_{\text{f,LL}} U_{\text{LL}} U_2 / 2 / m_f \quad (\text{A27})$$

$$U_{\text{LL}} = 0.816(2\Delta P_{\text{LL}} / \rho_2)^{1/2} \quad (\text{A28})$$

$$\Delta P_{\text{LL}} = m_f (D_2 C_{2u} / 2 - D_1 C_{1u} / 2) / Z_b / (D_1 / 4 + D_2 / 4) / (b_1 / 2 + b_2 / 2) / L_m \quad (\text{A29})$$

$$m_{\text{f,LL}} = \rho_2 Z_b \epsilon U_{\text{LL}} L_m \quad (\text{A30})$$

where $m_{\text{f,LL}}$ is the blade clearance leakage flow rate, and U_{LL} is the velocity of the clearance gap flow.

[Recommended articles](#)

Data availability

Data will be made available on request.

References

- [1] G.P. Thiel, A.K. Stark
To decarbonize industry, we must decarbonize heat
Joule, 5 (2021), pp. 531-550
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [2] Savut I (2019). Industrial Heat: Deep Decarbonization Opportunities (Bloomberg New Energy Finance).
[Google Scholar ↗](#)
- [3] Global Carbon Project (2019). Supplemental data of Global Carbon Budget 2019 (Version 1.0). Global Carbon Project.
<https://doi.org/10.18160/gcp-2019> ↗.
[Google Scholar ↗](#)
- [4] Roelofsen O, Somers K, Speelman E, Witteveen M (2020). Plugging in: What electrification can do for industry.
[Google Scholar ↗](#)
- [5] P. Sorknæs, R.M. Johannsen, A.D. Korberg, T.B. Nielsen, U.R. Petersen, B.V. Mathiesen
Electrification of the industrial sector in 100% renewable energy scenarios
Energy, 254 (2022), Article 124339
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [6] Hasanbeigi A, Springer C, Zuberi J, Lu HY, Zhou N (2019). Electrifying Industrial Heating in China.
[Google Scholar ↗](#)
- [7] G. Kosmadakis
Estimating the potential of industrial (high-temperature) heat pumps for exploiting waste heat in EU industries
Appl Therm Eng, 156 (2019), pp. 287-298
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [8] A. Gailani, S. Cooper, S. Allen, A. Pimm, P. Taylor, R. Gross
Assessing the potential of decarbonization options for industrial sectors

Joule, 8 (3) (2024), pp. 576-603



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[9]

J.T. Jiang, B. Hu, R.Z. Wang, N. Deng, F. Cao, C.C. Wang

A review and perspective on industry high-temperature heat pumps

Renew Sust Energ Rev, 161 (2022), Article 112106



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[10]

H.Z. Yan, C. Zhang, Z. Shao, M. Kraft, R.Z. Wang

The underestimated role of the heat pump in achieving China's goal of carbon neutrality by 2060

Engineering, 23 (2023), pp. 13-18



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[11]

K. Hamid, U. Sajjad, M.U. Ahrens, S. Ren, P. Ganesan, I. Tolstorebrov, *et al.*

Potential evaluation of integrated high temperature heat pumps: a review of recent advances

Appl Therm Eng, 230 (2023), Article 120720



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[12]

C. Arpagaus, F. Bless, M. Uhlmann, J. Schiffmann, S.S. Bertsch

High temperature heat pumps: market overview, state of the art, research status, refrigerants, and application potentials

Energy, 152 (2018), pp. 985-1010



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[13]

Dumont O, Frate GF, Pillai A, Lecompte S, De paepe M, Lemort V. Carnot battery technology: a state-of-the-art review. J

Energy Storage 2020; 32: 101756.

[Google Scholar ↗](#)

[14]

Y.L. Yan, F.B. Yang, H.G. Zhang, M.Z. Yu, X. Ping, Y.C. Pan, *et al.*

Thermodynamic evaluation of a novel Rankine-based pumped thermal energy storage concept targeting thermal coordination and large temperature span

Energ Convers Manage, 309 (2024), Article 118439



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [15] C. Mateu-Royo, C. Arpagaus, A. Mota-Babiloni, J. Navarro-Esbrí, S.S. Bertsch

Advanced high temperature heat pump configurations using low GWP refrigerants for industrial waste heat recovery: a comprehensive study

Energ Convers Manage, 229 (2021), Article 113752



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [16] Y.Y. Tan, Z.N. Hou, L. Wang, X.Z. Li, Z.W. Wang, J.F. Yuan

Performance assessment of a two-stage evaporation grade compression heat pump system for double temperature level drinking water production

Appl Therm Eng, 228 (2023), Article 120488



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [17] J. Gómez-Hernández, R. Grimes, J.V. Briongos, C. Marugán-Cruz, D. Santana

Carbon dioxide and acetone mixtures as refrigerants for industry heat pumps to supply temperature in the range 150–220 °C

Energy, 269 (2023), Article 126821



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [18] D. Wu, B. Hu, R.Z. Wang, H.B. Fan, R.J. Wang

The performance comparison of high temperature heat pump among R718 and other refrigerants

Renew Energ, 154 (2020), pp. 715-722



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [19] A. Vannoni, A. Sorce, A. Traverso, A.F. Massardo

Techno-economic optimization of high-temperature heat pumps for waste heat recovery

Energ Convers Manage, 290 (2023), Article 117194



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [20] R. Huang, B. Hu, R.Z. Wang, T.M. Eikevik, T.S. Ge
Thermodynamic and economic analysis of two-stage CO₂ Heat Pump with reverse Brayton cycle
Int J Refrig, 143 (2022), pp. 157-165
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [21] T.E. Samad, A. Żabnieńska-Góra, H. Jouhara, A.I. Sayma
A review of compressors for high temperature heat pumps
Therm Sci Eng Prog, 51 (2024), Article 102603
[Google Scholar ↗](#)
- [22] O. Bamigbetan, T.M. Eikevik, P. Neksa, M. Bantle
Review of vapour compression heat pumps for high temperature heating using natural working fluids
Int J Refrig, 80 (2017), pp. 197-211
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [23] A. Uusitalo, T. Turunen-Saaresti, J. Honkatukia, J. Tiainen, A. Jaatinen-Värri
Numerical analysis of working fluids for large scale centrifugal compressor driven cascade heat pumps upgrading waste heat
Appl Energy, 269 (2020), Article 115056
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [24] IEA Technology Collaboration Programme on Heat Pumping Technologies, High temperature heat pump: Annex 58, task 1: Technologies – state of the art and ongoing developments for systems and components, 2022, accesible: <https://heatpumpingtechnologies.org/annex58/task1/>.
[Google Scholar ↗](#)
- [25] J. Schiffmann, D. Favrat
Experimental investigation of a direct driven radial compressor for domestic heat pumps
Int J Refrig, 32 (8) (2009), pp. 1918-1928
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [26] J. Schiffmann, D. Favrat
Design, experimental investigation and multi-objective optimization of a small-scale radial compressor for heat pump applications
Energy, 35 (2010), pp. 436-450
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [27] T. Turunen-Saaresti, P. Røyttä, J. Honkatukia, J. Backman
Predicting off-design range and performance of refrigeration cycle with two-stage centrifugal compressor and flash intercooler
Int J Refrig, 33 (2010), pp. 1152-1160
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [28] P. Røyttä, T. Turunen-Saaresti, J. Honkatukia
Optimising the refrigeration cycle with a two-stage centrifugal compressor and a flash intercooler
Int J Refrig, 32 (2009), pp. 1366-1375
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [29] A. Grauberger, D. Young, T. Bandhauser
Experimental validation of an organic Rankine-vapor compression cooling cycle using low GWP refrigerant R1234ze(E)
Appl Energy, 307 (2022), Article 118242
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [30] B. Hu, H. Liu, R.Z. Wanga, H.B. Li, Z.P. Zhang, S. Wang
A high-efficient centrifugal heat pump with industrial waste heat recovery for district heating
Appl Therm Eng, 125 (2017), pp. 359-365
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [31] C. Arpagaus, F. Bless, S.S. Bertsch
Theoretical analysis of transcritical HTHP cycles with low GWP HFO refrigerants and hydrocarbons for process heat up to 200 °C

Refrig Sci Technol (2020), pp. 342-350

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [32] A.Y. Sulaiman, D. Cotter, C. Wilson, B. Kamkari, N. Hewitt
Energetic and exergo-environmental analysis of transcritical high-temperature heat pumps with low GWP refrigerants for industrial waste heat recovery

Int J Refrig, 156 (2023), pp. 12-28



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [33] M. Casey, C. Robinson, Radial Flow Turbocompressors, Cambridge University Press, ISBN: 978-1-108-41667-2, 2021.

[Google Scholar ↗](#)

- [34] P.M. Came, C.J. Robinson
Centrifugal compressor design

Proc Inst Mech Eng C, 213 (2) (1999), pp. 139-155

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [35] G.F. Frate, L. Ferrari, U. Desideri
Analysis of suitability ranges of high temperature heat pump working fluids

Appl Therm Eng, 150 (2019), pp. 628-640



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [36] A. Meroni, B. Zuhlsdorf, B. Elmegaard, F. Haglind
Design of centrifugal compressors for heat pump systems

Appl Energy, 232 (2018), pp. 139-156



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [37] A. Jaatinen-Värri, J. Honkatukia, A. Uusitalo, T. Turunen-Saaresti
Centrifugal compressor design for high-temperature heat pumps

Appl Therm Eng, 239 (2024), Article 122087



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [38] A. Uusitalo, A. Jaatinen-Värri, T. Turunen-Saaresti, J. Honkatukia, J. Tiainen
Centrifugal compressor design and cycle analysis of large-scale high temperature heat pumps using hydrocarbons
Appl Therm Eng, 247 (2024), Article 123035
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [39] P. Jiang, Y. Tian, B. Wang, C.H. Guo
Design and analysis of centrifugal compressor in carbon dioxide heat pump system
Sci Rep, 14 (2024), p. 5286
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [40] T. Liang, A. Vecchi, K. Knobloch, A. Sciacovelli, K. Engelbrecht, Y.L. Li, *et al.*
Key components for Carnot Battery: Technology review, technical barriers and selection criteria
Renew Sust Energ Rev, 163 (2022), Article 112478
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [41] R.H. Aungier
Centrifugal compressors: a strategy for aerodynamic design and analysis
ASME Press, New York, NY, USA (2000)
[Google Scholar ↗](#)
- [42] H. Li, Y.P. Ju, C.H. Zhang
Optimization of supercritical carbon dioxide recompression Brayton cycle considering anti-condensation design of centrifugal compressor
Energy Convers Manage, 254 (2022), Article 115207
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [43] L.C. Yao, Z.P. Zou
A one-dimensional design methodology for supercritical carbon dioxide Brayton cycles: Integration of cycle conceptual design and components preliminary design
Appl Energy, 276 (2020), Article 115354

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [44] A. Giuffre, P. Colonna, M. Pini
The effect of size and working fluid on the multi-objective design of high-speed centrifugal compressors
Int J Refrig, 143 (2022), pp. 43-56
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [45] A. Giuffre, M. Pini
Design guidelines for axial turbines operating with non-Ideal compressible flows
J Eng Gas Turb Power, 143 (1) (2021), Article 011004
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [46] Lemmon EW, Huber ML, McLinden MO. NIST reference fluid thermodynamic and transport properties–REFPROP; 2002.
[Google Scholar ↗](#)
- [47] Y.S. Li, G.L. Lu
Centripetal turbine and centrifugal compressor
China Machine Press (1987)
[Google Scholar ↗](#)
- [48] L. Bourabia, S. Khalfallah, M. Cerdoun, T. Chettibi
An efficient methodology to generate optimal inputs for the preliminary design of centrifugal compressor impellers
P I Mech Eng E-J Pro, 234 (4) (2020), pp. 353-366
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [49] A. Whitfield
Preliminary design and performance prediction techniques for centrifugal compressors
Proc IMechE, Part A: J Power Energy, 204 (1990), pp. 131-144
[Google Scholar ↗](#)
- [50] C. Xu, R.S. Amano

Empirical design considerations for industrial centrifugal compressors

Int J Rotat Mach, 2012 (2012), pp. 1-15

[Crossref ↗](#) [Google Scholar ↗](#)

- [51] A. Perdichizzi, M. Savini
Aerodynamic and geometric optimization for the design of centrifugal compressors
Int J Heat & Fluid Flow, 6 (1) (1985), pp. 49-56
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [52] Y.H. Xu, X. Wang, J. Zhang, *et al.*
Experimental investigation and artificial neural network prediction of small-scale compressed air energy storage system based on pneumatic motor
Therm Sci Eng Prog, 47 (2024), Article 102287
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [53] Wang YF. Researches on centrifugal compressor for Transcritical CO₂ heat pump. China; Chinese Academy of Sciences, 2021.
[Google Scholar ↗](#)
- [54] X.H. Yu, H.N. Qiao, B. Yang, H.T. Zhang
Thermal-economic and sensitivity analysis of different Rankine-based Carnot battery configurations for energy storage
Energ Convers Manage, 283 (2023), Article 116959
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [55] A.Y. Sulaiman, D. Cotter, C. Arpagaus, N. Hewitt
Theoretical evaluation of energy, exergy, and minimum superheat in a high-temperature heat pump with low GWP refrigerants
Int J Refrig, 153 (2023), pp. 99-109
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [56] M. Galvas
Fortran program for predicting off-design performance of centrifugal compressors
Tech Rep (1973)
[Google Scholar ↗](#)
- [57] J. Coppage, F. Dallenbach
Study of supersonic radial compressors for refrigeration and pressurization systems
Tech Rep, Garret Corp (1956)
[Google Scholar ↗](#)
- [58] W. Jansen
A method for calculating the flow in a centrifugal impeller when entropy gradients are present
Roy Soc Conf Internal Aerodyn (Turbomach) (1967), pp. 133-146
[Google Scholar ↗](#)
- [59] R.H. Aungier
Mean streamline aerodynamic performance analysis of centrifugal compressors
Trans ASME – J Turbomach, 117 (1995), pp. 360-366
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [60] J. Hosseinpour, M. Messele, A. Engeda
Design and development of a stable supercritical CO₂ centrifugal compressor
Therm Sci Eng Prog, 47 (2024), Article 102273
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [61] J.W. Daily, R.E. Nece
Chamber dimension effects on induced flow and frictional resistance of enclosed rotating disks
Trans ASME, Journal of Basic Engineering, 82 (1960), pp. 217-232
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [62] H.W. Oh, E.S. Yoon, M.K. Chung

An optimum set of loss models for performance prediction of centrifugal compressors

Proc Inst Mech Eng, Part A: J Power Energy, 211 (4) (1997), pp. 331-338

[View in Scopus ↗](#) [Google Scholar ↗](#)

Cited by (4)

[A multi-objective optimization framework integrating DNN-SHAP and NSGA III-VIKOR for isentropic efficiency and surge margin enhancement of a centrifugal impeller](#)

2026, Aerospace Science and Technology

[Show abstract](#) ✓

[A rapid and cost-effective optimization framework for supercritical CO2 centrifugal compressor blades using a hybrid prediction-CFD approach](#)

2026, Energy

[Show abstract](#) ✓

[Assessment of low-GWP refrigerants in ultra-high-temperature transcritical heat pumps: a systematic evaluation framework based on life-cycle economics](#)

2026, Applied Thermal Engineering

[Show abstract](#) ✓

[Investigation of energy dissipation mechanisms and unsteady flow characteristics of a cooling pump based on sweeping jet inflow](#)

2026, Energy Conversion and Management

[Show abstract](#) ✓

1 The two authors have the same contribution to this study.

[View Abstract](#)

© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.